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Identification of Motor Areas of the Cat Cerebral Cortex Based on Studies of Cortical Stimulation and Corticospinal Connections

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ABSTRACT

The location and topography of motor areas in the cat cerebral cortex were studied by electrical stimulation of the cortex in five animals, and by the injection of retrograde tracers into the spinal cord of four animals. Movements evoked by intracortical microstimulation (ICMS) of the anterior, posterior and lateral sigmoid gyri, both banks of the cruciate sulcus and the dorsal bank of the presylvian sulcus were observed in anaesthetized cats. Fluorescent tracers (Fast Blue and/or Diamadino Yellow) were injected into the lateral funiculus in the second cervical segment, into the gray matter of cervical segments C3–T1 and/or into the gray matter of lumbar segments L2–S1.

Contraction of the contralateral forelimb, hindlimb or facial muscles was observed following electrical stimulation of several cytoarchitectonic areas: 4γ, 4δ, 6aα, 6aγ, and 3a. These findings suggested representations of contralateral forelimb and hindlimb movements in areas 4γ and 4δ, and of the contralateral forelimb muscles in areas 6aα and 6aγ. Corticospinal neurons were located in all the above cytoarchitectonic areas as well as in areas 3b, 1, 2, 2pri, and 5. Large numbers of neurons were labeled in areas 4γ and 4δ, and moderate labeling was observed in areas 6aγ and 6aα. Corticospinal neurons projecting to cervical and lumbar segments were located in areas 4γ and 4δ, while those projecting only to cervical segments were detected in areas 6aα and 6aγ.

Based on these findings it is proposed that within the motor cortex of the cat there are representations of limb movements in several cytoarchitectonic subdivisions. Many of these representations may be candidate secondary motor areas. J. Comp. Neurol. 380:191–214, 1997. © 1997 Wiley-Liss, Inc.

Indexing terms: intracortical microstimulation; fluorescent tracers; primary motor area; secondary motor area

The cortical stimulation studies of Woolsey (1958) were interpreted to recognize, in a number of different mammals, at least two major motor areas controlling limb movements: the primary motor area (MI) and the supplementary motor area (SMA or MII). Whether these two areas are homologous across species, however, continues to be debated (see Wise, 1996). Primates, particularly macaques, have been extensively studied in this regard, and a lateral premotor area (PM) and a cingulate motor area (CMA) have also been identified in these animals (reviewed by Wise et al., 1991; Dum and Strick, 1991; Kurata, 1994a). A variety of anatomical and physiological techniques, including cortical stimulation studies, and studies of corticospinal and corticocortical connections have been used to identify motor areas of the cortex (reviewed by Humphrey, 1986; Wise et al., 1991; Dum and Strick, 1991; Porter and Lemon, 1993; Wise, 1996).

Carnivores, particularly cats, are widely used for the experimental study of cortical function, including motor function. Although there have been many studies that have examined the cytoarchitecture (Hassler and Muhs-Clements, 1964), motor responses to stimulation (Delgado,

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1952; Delgado and Livingston, 1955; Asanuma and Ward, 1971; Nieoullon and Rispal-Padel, 1976; Pappas and Strick, 1981; Keller, 1993), and corticospinal and corticocortical connections (Armand et al., 1974; Groos et al., 1978; Reinoso-Suarez, 1984; Yumiya and Ghez, 1984) of the cat motor cortex, many questions remain to be answered. There appears to be enough data in the literature to locate the primary motor area in area 4γ (see Avendano et al., 1992), but the "secondary" motor areas (e.g., SMA, PM) have not been well characterized in the cat. Data regarding the location of secondary motor areas in other carnivores (e.g., dog, raccoon) are also meager (Jameson et al., 1968; Gorska, 1974).

Many difficulties in the attempts to characterize "secondary" motor areas in the cat may be traced to differences in organization between cat and primate motor areas. For example, a more or less complete representation of contralateral body parts has been demonstrated in area 4γ but not in any other parts of area 4 or 6 in the cat (Delgado, 1952; Delgado and Livingston, 1955; Nieoullon and Rispal-Padel, 1976; Keller, 1993). Corticospinal neurons are sparse or absent in cytoarchitectonic area 6 (Armand et al., 1974; Groos et al., 1978) unlike that demonstrated in the primate (Toyoshima and Sakai, 1982; He et al., 1993, 1995; Galea and Darian-Smith, 1994).

Since area 4 of the cat cortex is divisible cytoarchitectonically into several subregions (Hassler and Muhs-Clements, 1964), it is possible that parts of area 4, particularly area 4δ, may qualify as candidate "secondary" motor areas. Other studies indicate that area 6 may have projections to the spinal cord (Hayes and Rustioni, 1981; Catsman-Berrevoets and Kuypers, 1981; Nudo and Masterton, 1990) and electrical stimulation of area 6 may evoke movements (Delgado, 1952; Delgado and Livingston, 1955; Nieoullon and Rispal-Padel, 1976). In order to define secondary motor areas in the cat, we have reexamined area 4 and area 6 of the cat cortex by intracortical microstimulation studies, and tracer studies of corticospinal connections. The results suggest that parts of area 6 and 4δ project to the spinal cord, have representations of contralateral body parts, and may qualify as secondary motor areas.

Abbreviations

Cerebral sulci

SA	ansate sulcus
dSA, fSA, vSA	dorsal lip, floor and ventral lip of ansate sulcus (the above prefixes apply to maps of all unfolded sulci)
SCor	coronal sulcus
SCr	cruciate sulcus
SEsA	anterior ectosylvian sulcus
SL	lateral sulcus
SO	orbital sulcus
SPs	presylvian sulcus
SRA	anterior rhinal sulcus
SSp	splenial sulcus
SSs	suprasylvian sulcus

Cytoarchitectonic areas of the cerebral cortex

r4γ	rostral subdivision of area 4γ
c4γ	caudal subdivision of area 4γ
r4δ	rostral subdivision of area 4δ
c4δ	caudal subdivision of area 4δ
Ins	insula
1, 2, 3a, 3b, 5, 7, 2pri	other subdivisions

Joints moved by intracortical microstimuli

An	ankle
E	elbow
F	face
H	hip
Kn	knee
Sh	shoulder
Wr	wrist

MATERIALS AND METHODS

Experiments were conducted on nine adult cats of either sex weighing between 2.2 kg and 4 kg. Five animals were used for cortical stimulation studies while retrograde tracers were injected into the spinal cord of the other four animals. All experimental protocols were approved by the University of Queensland Animal Experimentation Ethics Committee.

Cortical stimulation

Animals were anesthetized with ketamine (5 mg/kg i.m.) and xylazine (0.05 mg/kg i.m.). Subsequently, light anesthesia was maintained with i.v. Ketamine (3 mg/kg) and pentobarbitone (2 mg/kg) based on the appearance of the withdrawal reflex. Atropine (0.06 mg/kg, s.c) and dexamethazone (0.5 mg/kg, i.m.) was administered at the beginning of the experiment. Blood pressure, expired CO_2 concentration and rectal temperature were monitored throughout the experiment. A feedback controlled electric blanket was used to keep the animal warm (37.5 ± 1°C) and intravenous fluids (isotonic saline) administered periodically.

The head was fitted in a sterotaxic frame and the thorax and abdomen supported with foam cushions in order to allow the limbs to hang loosely by the side. A craniotomy was performed on the left side in order to expose the frontal lobe. The dura was reflected over the pericruciate region and the cortical surface kept moist with warm normal saline. Glass-coated tungsten electrodes (Re = 0.5MΩ) were inserted into the cortex and a train of cathodal pulses (16 pulses, each 0.2 msec wide, frequency 400 Hz) was delivered at depths from 1.5 mm to 11.0 mm at 1 or 1.5 mm intervals. The stimulus current was monitored and varied from 1 to 100 μA. The cortex was sampled in a grid-like manner at intervals of 1 mm (avoiding surface vessels) and penetrations were extended posteriorly up to the caudal end of the ansate sulcus in order to reach the depths of the cruciate sulcus. Lesions were frequently made (by passing 10 μA for 10 seconds) when movements were evoked from cortical regions deeper than 3–4 mm. At least one lesion was made in most successful tracts. However, sometimes we avoided making too many closely spaced lesions by not lesioning in adjacent penetrations at similar depths.

At least two observers were present throughout each experiment in order to identify the movements evoked by intracortical microstimulation (ICMS). Observation and palpation of muscle contraction was used to identify evoked movements. Limbs were repositioned if initial trials were unsuccessful. Current thresholds for evoking movements were also determined by both observers.

Tracer injections

The fluorescent tracers, Fast Blue (FB, Sigma) and Diamadino Yellow (DY, Sigma) were injected into the spinal cord of four animals (cases 6 to 9). Anesthesia was induced by ketamine and xylazine and maintained with ketamine and pentobarbitone administered intramuscularly (see above for dosages). Atropine and steroid were administered preoperatively (see above). The rectal temperature was monitored and the animal kept warm using a

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TABLE 1. Volume and Concentration of Tracers Injected in the Spinal Cord

Animal case no.	Tracer and conc.	Volume (μl)	Penetrations	Location
6	FB[1] (5% sol.)	2.5	1	Lateral funiculus
7	DY[1] (2% sol.)	9.2	20	C3–T1 gray
8	FB (5% sol.)	6.0	12	L2–S1 gray
9	DY (2% sol.)	10.0	20	C5–T1 gray
	FB (5% sol.)	5.0	10	L2–S1 gray

[1]FB, Fast Blue; DY, Diamadino Yellow.

feedback controlled electric blanket. The surgery was performed under aseptic conditions and antibiotics, analgesics and steroids injected postoperatively.

The head and pelvis of the animal were fixed in a stereotaxic frame and the vertebral spines above and below the desired laminectomy were fixed with clamps. A laminectomy was then performed to expose the following spinal segments: C2, C3 in case 6, C3–T1 in case 7 and L2–S1 in case 8. In case 9, two laminectomies were performed separated by 3 weeks. The first laminectomy involved L2 to S1 spinal segements, while the second involved C5–T1 segments. Hamilton syringes (1 μl) were used to inject tracer into the lateral funiculus (case 6) or spinal gray matter (cases 7–9) (Table 1, Fig. 1). After a single penetration into the lateral funiculus of C2 segment (case 6, Fig. 1) 2.5 μl of tracer (FB) was injected about 3 mm below the surface. Multiple penetrations, spaced about 5 mm apart, were made into the spinal cord of cases 7–9 to involve the desired spinal segments (Table 1, Fig. 1). About 0.5 μl of tracer was injected at each penetration at a depth of about 3.5 mm in order to involve the intermediate zone (and adjoining parts of the ventral and dorsal horn). The penetrations were made vertically, midway between the midline and the dorsal root entry zone, through the dorsal columns (Fig. 1). The spinal segments were identified by counting the dorsal roots below the twelfth thoracic vertebra or second cervical vertebra. Approximately 2–4 weeks following the tracer injection in the cervical or lumbar spinal cord, respectively, the animal was reanesthetized for perfusion.

Perfusion and histological procedures

At the end of the ICMS experiments or after appropriate survival times (2–4 weeks after tracer injection in the spinal cord) the animal was deeply anaesthetized (i.v. pentobarbitone 50 mg/kg) and perfused with 2 liters of normal saline followed by 2 liters of 4% paraformaldehyde in phosphate buffer. The brain was removed, photographed and soaked in 30% sucrose. The spinal cord was removed from animals with tracer injections.

Each hemisphere was cut serially on a freezing microtome (50 μm). Brain sections from cats subjected to cortical stimulation were (every alternate 50 μm section) collected and mounted onto slides. The sections were stained with cresyl violet or thionine, dehydrated, cleared in xylene and coverslipped. The sections were then examined for electrode tracts and lesions (Fig. 2).

Brain and spinal cord (appropriate segments) sections from cats with tracer injections were divided into four batches. The first batch was mounted on slides, dehydrated, cleared and coverslipped for examination of fluorescent label. The second batch was stained with cresyl violet or thionine for cytoarchitectonic study.

Identification of injection sites and labeled cortical neurons

Cresyl violet-stained sections were drawn at a magnification of 44 through a binocular microscope (Zeiss Axioskop) with a drawing tube (objective ×10, drawing tube ×4.4, Fig. 3). The unstained section was then projected to overlap the drawing accurately, and labeled neuronal profiles were identified and plotted on the drawing. The labeled profiles were then observed under higher power (objective ×20, ×50) to confirm plotted points or identify double labeled neurons (Fig. 3). The injection sites in the spinal cord were similarly documented. Cytoarchitectonic borders of the cerebral cortex were included in the drawing.

Identification of cytoarchitectonic boundaries

Cytoarchitecture of the motor and somatosensory areas of the cerebral cortex was based on Hassler and Muhs-Clements (1964; Figs. 4–7). Subsequently, other studies (Avendano and Verdu, 1992; Avendano et al., 1992; LeClerc et al., 1994; Myasnikov et al., 1994) have elaborated and modified some of these criteria. One of the major changes proposed by Avendano et al. (1992) was the subdivision of area 6aβ into a laterally located 6aγ and a medially located 6aβ. Further, the boundary of area 6aα was extended rostrally, so that it occupied a portion of the cortex in the anterior sygmoid gyrus. We followed the newer descriptions of Avendano et al. (1992) because they also distinguished parts of area 6 with and without corticospinal projections.

Hassler and Muhs-Clements (1964) subdivided area 4 into four cytoarchtectonic areas: 4γ, 4δ, 4 suprafundum (4sfu) and 4 in fundo (4fu). Of these, only 4γ contains large numbers of giant pyramidal cells, often arranged in multiple rows. The width of area 4δ cortex is smaller and giant pyramidal cells are less frequent (Fig. 4). The cortical width in areas 4sfu and 4fu decreases further and the pyramidal cells in layer V are even smaller. The overall thickness of the cortex diminishes gradually with increasing depth in the cruciate sulcus and at the border of areas 4γ and 4δ the thickness of these two areas may be similar (Fig. 2) although the size and clustering of pyramidal cells in lamina V still allow distinction of the two areas. Differences in the width and cellular arrangement in lamina III also distinguish the different subdivisions of area 4 (Hassler and Muhs-Clements, 1964). The exact borders between areas 4δ, 4fu and 4sfu are not always easy to define but the illustrated margins are within 50–200 μm of clearly distinguishable cytoarchitecture.

Hassler and Muhs-Clements (1964) distinguished three subdivisions within area 6: area 6aα, 6aβ and 6iffu. Area 6aβ (as defined by Hassler and Muhs-Clements, 1964) involved a wide region of cortex including the medial surface of the cortex (below the cruciate sulcus), a part of the rostral or ventral bank of the cruciate sulcus, the anterior sygmoid gyrus and the dorsal bank of the presylvian sulcus. Subsequently, Avendano et al. (1992) have pointed out cytoarchitectonic variations within area 6aβ, which were also confirmed in our study. We found that area 6aβ is confined to a thin strip of cortex in the medial part of the anterior sigmoid gyrus, and a thin strip of cortex adjoining the cruciate sulcus on the medial surface and in the rostral (ventral) bank of the cruciate sulcus. Layer V in

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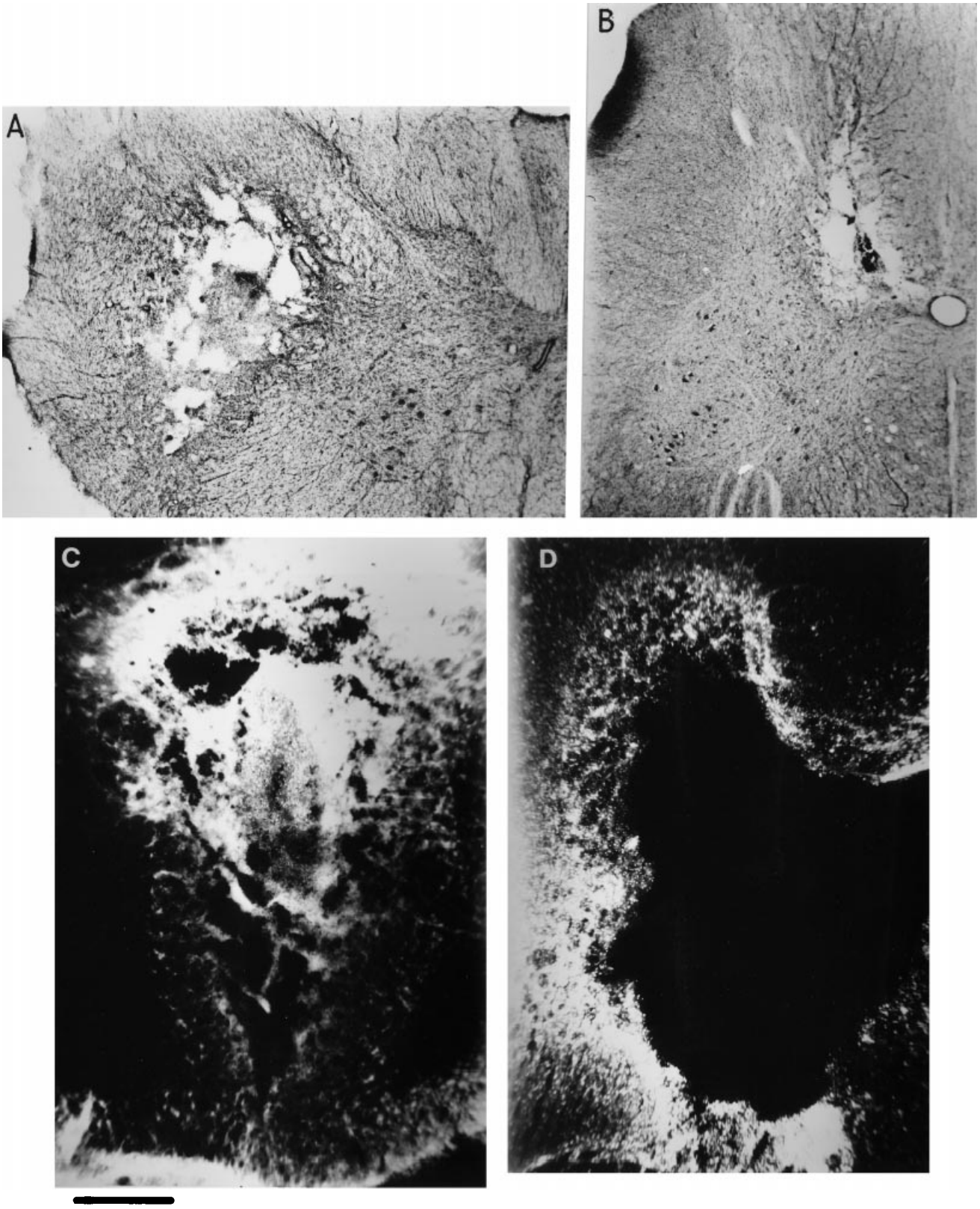


Fig. 1. Photomicrographs of tracer injection sites in the spinal cord of cases 6 (**A, C**) and 7 (**B, D**). A and B illustrate the needle tracts in the right lateral column (white matter) of case 6 and in the right intermediate gray matter of case 7 in lightly stained cross-sections of the cervical spinal cord. C and D illustrate the Fast Blue and Diamadino Yellow injection sites, respectively, in these animals. Calibration bar = 0.8 mm in A, B; 0.5 mm in C, D.

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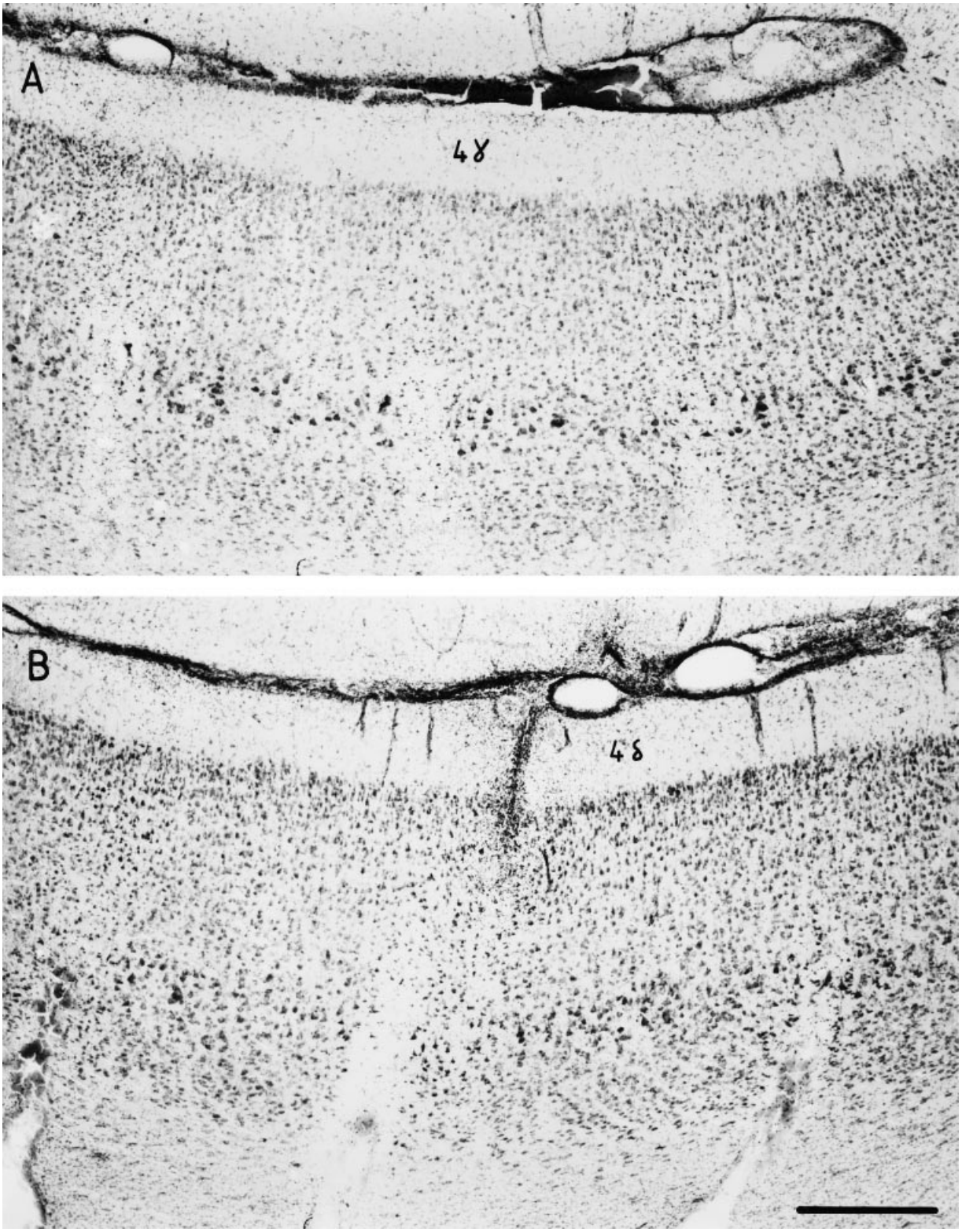


Fig. 2. Photomicrographs of coronal sections (stained with cresyl violet) of the brain of a cat used for the cortical stimulation study, illustrating electrode tracts in the dorsal (caudal) bank of the crucite sulcus in areas 4γ (**A**) and 4δ (**B**). In this illustration dorsal is down and medial is to the left. Calibration bar = 0.5 mm.

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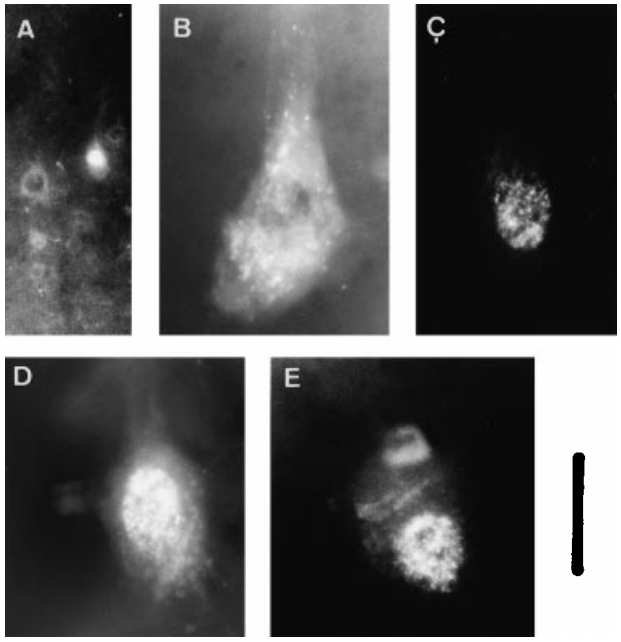


Fig. 3. Photomicrograph of corticospinal neurons labeled in case 9, illustrating neurons labeled by Fast Blue (**A, B**), Diamadino Yellow (**C**), and both tracers (double labeled, A, **D, E**). Calibration bar = 100 μm in A; 20 μm in B–E.

this region has medium-sized pyramidal cells (Fig. 5). In contrast, area 6aγ, located in the dorsal bank of the presylvian sulcus and at its rostral end, has larger pyramidal cells, including many giant pyramidal neurons in layer V (see Avendano et al., 1992 and Fig. 6). Area 6aα also has large pyramidal cells in layer V (and a few giant pyramidal cells) and extends into the anterior sigmoid gyrus. The border between areas 6aβ and 6aα is easily distinguishable: a more clearly defined laminar arrangement and the organization of palisades in lamina VI characterize area 6aβ (Fig. 5). Area 6aγ has larger pyramidal cells in lamina V (than area 6aα), including giant pyramidal neurons, and lamina VI is wider (Fig. 6). Area 6iffu located deep in the rostral (ventral) bank of the cruciate sulcus has a much thinner cortex, and lamina V is narrow with few large pyramidal cells (Hassler and Muhs-Clements, 1964; Fig. 6). The exact border between area 6aα and 6iffu may be difficult to define in the coronal series (±100 μm).

Area 3a is distinguishable from area 4γ by the presence of a thin lamina IV, and an overall reduction in cortical thickness, but the latter property is not always evident (Fig. 7). Large pyramidal cells are frequently located in lamina V of area 3a (Avendano and Verdu, 1992). Area 3b has a thick and prominent lamina IV (Fig. 7). In lateral cortical regions, lamina V is light with no large pyramidal cells (Fig. 7), but more medially, large pyramidal cells may be frequently seen (LeClerc et al., 1994).

RESULTS

Electrical stimulation of the motor cortex

Intracortical microstimuli were delivered to the anterior, posterior and lateral sigmoid gyri. In addition, the banks of the cruciate sulcus and the dorsal bank of the presylvian sulcus were explored. The dorsal surface of the cortex was explored in a grid-like manner at intervals of 1 mm between adjacent penetrations, avoiding surface vessels. Electrode tracts and lesions were reconstructed from histological sections in order to locate the area stimulated (Fig. 8). The stimulated points and the joints moved were then mapped on the cortical surface and in the banks of the cruciate and presylvian sulci (Figs. 9–11). The electrode tracts were sometimes found to deviate from the vertical in the depths of the cortex (Fig. 8). Therefore, the exploration of the depths of the cruciate and presylvian sulci was not uniform (Figs. 9, 10).

The cortical surface was reconstructed from coronal sections (in the plane of the upper border of lamina V or lower border of lamina IV) as follows. The medial edge of the hemisphere was used as the reference to align all sections and the distances between this edge and various sulci and the inferior edge of the hemisphere were measured and plotted. The points were then joined using a smooth line. The banks of the presylvian sulcus and cruciate sulcus were unfolded separately at the upper border of lamina V. The cytoarchitectonic boundaries were then included in the drawing.

The ICMS maps from cases 1 and 2 are presented in Figures 9 and 10, respectively, while Figure 11 represents results from all five animals studied. ICMS was effective in evoking movements from areas 4γ, 4δ, 6aα, 6aγ, and area 3a. Although stimuli in areas 1 and 2 were also effective, these areas were not studied in detail; hence, data are not included. Electrical stimuli were ineffective in areas 6aβ, 6iffu and 4sfu. Within the areas from which movements were successfully evoked, many stimulated points were ineffective. In such situations, trials were repeated with the limbs repositioned and after a short delay (5–10 minutes), success was obtained at some points.

Stimulation of the banks of the cruciate sulcus or presylvian sulcus was usually only effective when the stimulated point was in lamina V or VI (Fig. 8). Stimuli delivered at more superficial cortical laminae rarely evoked movements, required high currents, and were not included in illustrations or in analysis of current thresholds. Stimulus points located in white matter were not analyzed.

Movements were frequently evoked contralaterally in the forelimb or hindlimb. These were invariably flexion movements, though occasionally pronation and supination movements appeared to be superimposed on wrist flexion. Similarly, abduction and adduction, in addition to flexion, was sometimes observed at the shoulder or hip joints. Facial movements involved the contralateral eyelids, cheeks or lips; jaw movements were not observed. Since facial movements were infrequently evoked in our study, the location of the cortical representation of facial muscle must be cautiously interpreted. Forelimb movements were evoked from areas 4γ, 4δ, 6aα and 6aγ, hindlimb movements from areas 4γ and 4δ and facial movements from areas 4γ, 6aα and 6aγ.

In dealing with somatotopic organization within these areas, we decided to subdivide areas 4γ and 4δ into rostral (r4γ, r4γ) and caudal (c4γ, c4δ) sectors, separated by the cruciate sulcus (or its fundus) for several reasons. First, it has been suggested that electrical thresholds for evoking movements are lower in r4γ than c4γ (Nieoullon and Rispal-Padel, 1976). Secondly, separate representations of the forelimb and hindlimb in r4γ and c4γ are suggested by the study of Groos et al. (1978). Thirdly it has been shown

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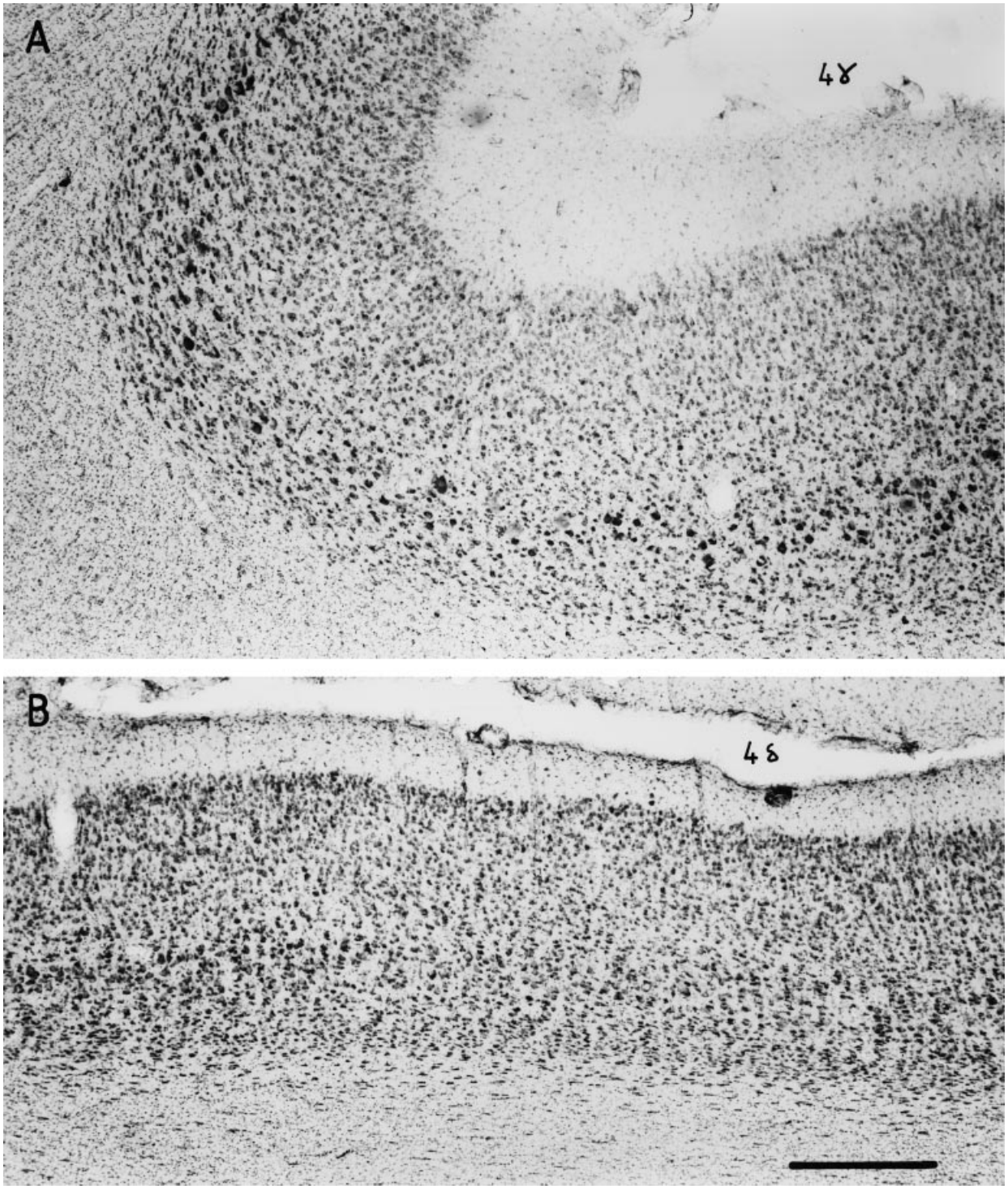


Fig. 4. Photomicrographs of coronal sections of the cat cerebral cortex, stained with cresyl violet, illustrating the cytoarchitecture of areas 4γ (**A**) and 4δ (**B**), in the dorsal (caudal) bank of the cruciate sulcus. Dorsal is down and medial is to the right. Calibration bar = 0.5 mm.

that rostral and caudal 4γ have different cortical connections and functional specializations (Adkins et al., 1971; Vicario et al., 1983; Yumiya and Ghez, 1984). Finally, cortical mapping studies in dogs and racoons suggest that there may be separate representations of contralateral body parts in the anterior and posterior sigmoid gyri (Hardin et al., 1968; Jameson et al., 1968; Gorska, 1974).

The study found representations of forelimb and hindlimb movements in several areas including r4γ, c4γ, r4δ and c4δ. Stimulation of area 6aα and 6aγ resulted in

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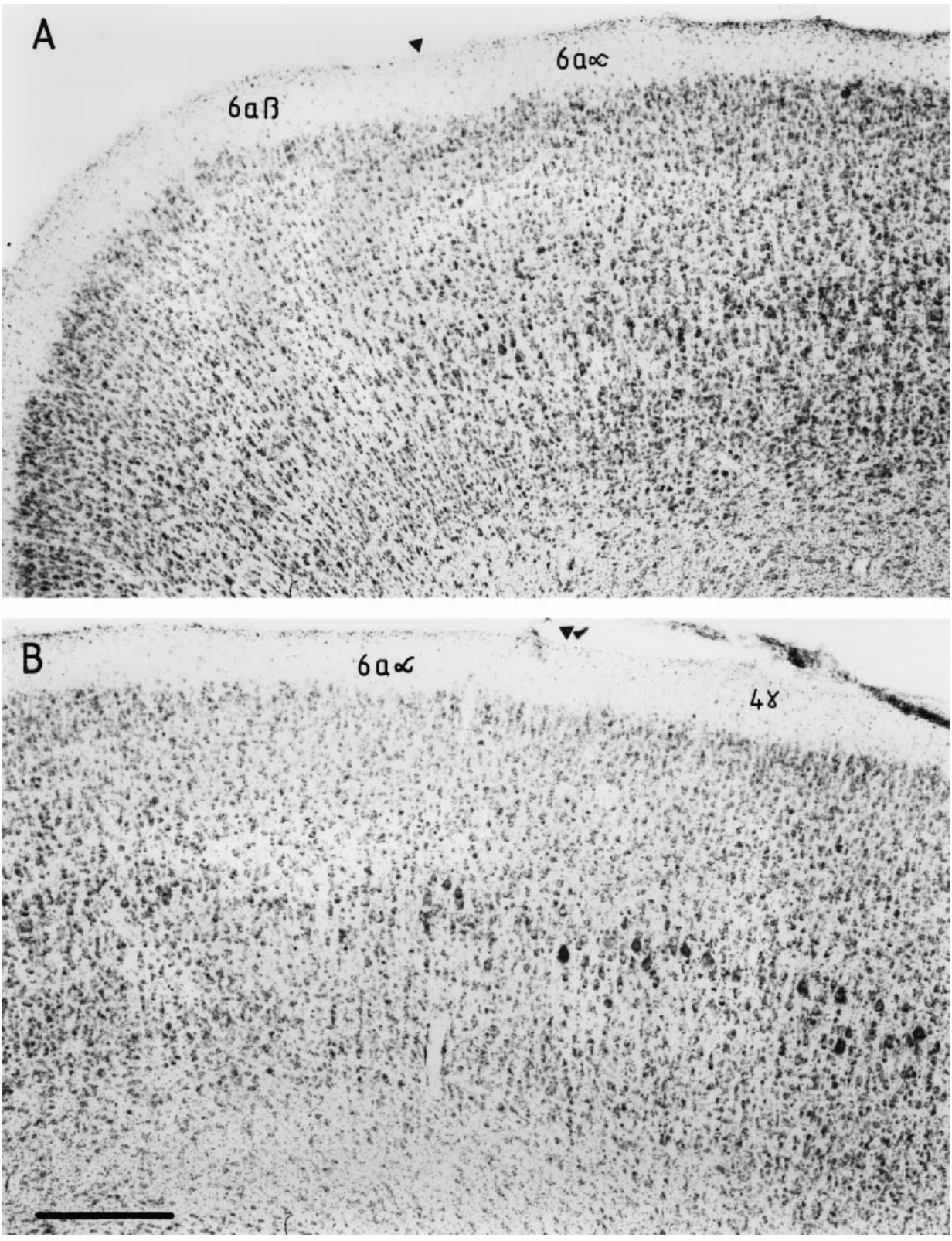


Fig. 5. Photomicrographs of coronal sections of the cat cerebral cortex, stained with cresyl violet, contrasting the cytoarchitecture of areas 6aβ and 6aα (**A,** the border between 6aβ, left, and 6aα, right, is indicated by the arrowhead above the pia), and 6aα and 4γ (**B,** the border between 6aα, left, and 4γ, right, is indicated by the arrowhead above the pia), in the ventral (rostral) bank of the cruciate sulcus. Dorsal is up and medial is to the left. Calibration bar = 0.5 mm.

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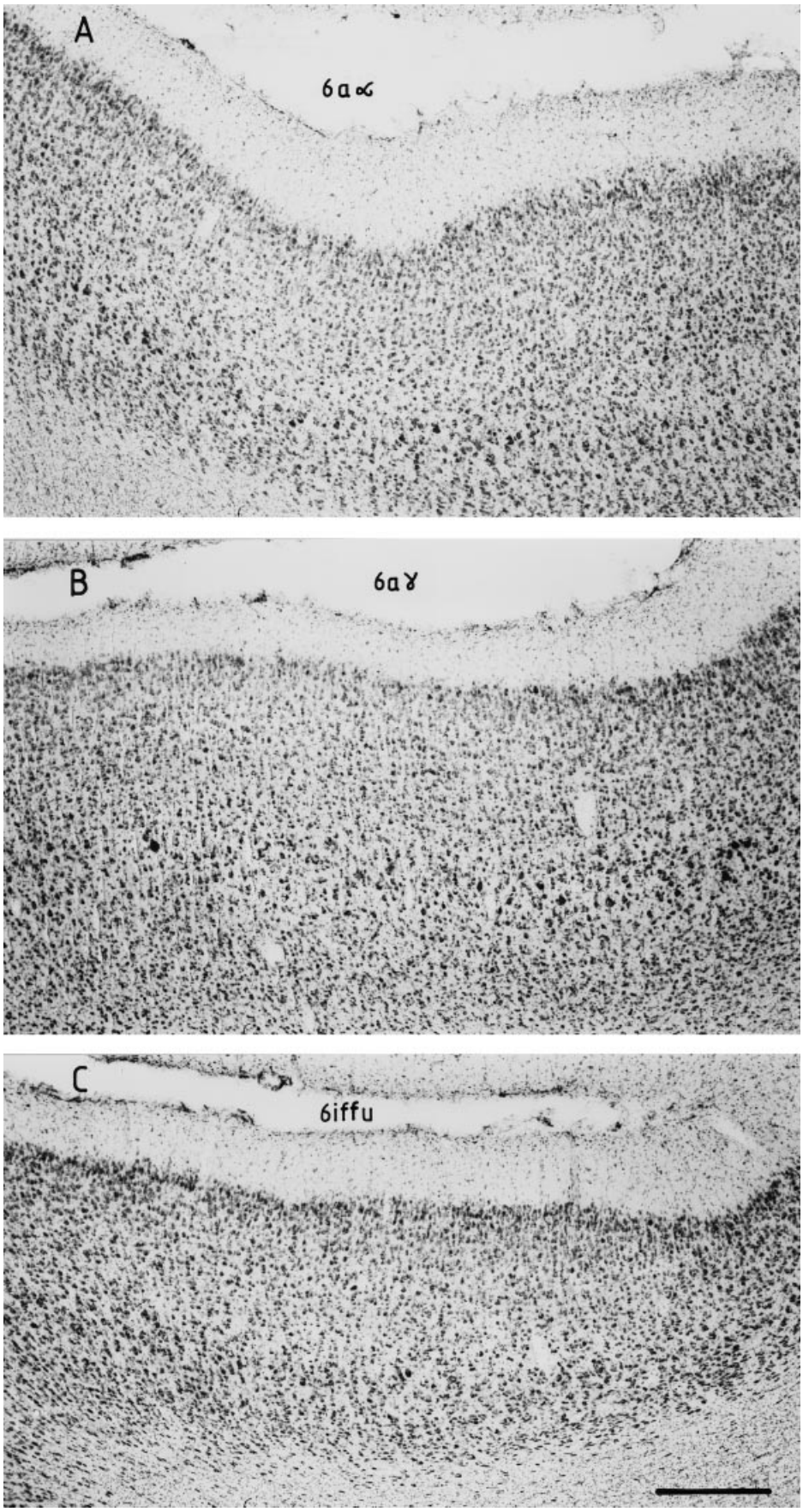


Fig. 6. Photomicrographs of coronal sections of the cat cerebral cortex, stained with cresyl violet, illustrating the cytoarchitecture of area 6aα in the ventral/rostral bank of the cruciate sulcus (**A**), area 6aγ in the dorsal bank of the presylvian sulcus (**B**), and area 6iffu in the ventral/rostral bank of the cruciate sulcus (**C**). In A and C, dorsal is up and medial is to the left, while in B dorsal is down and medial is to the right. Calibration bar = 0.5 mm.

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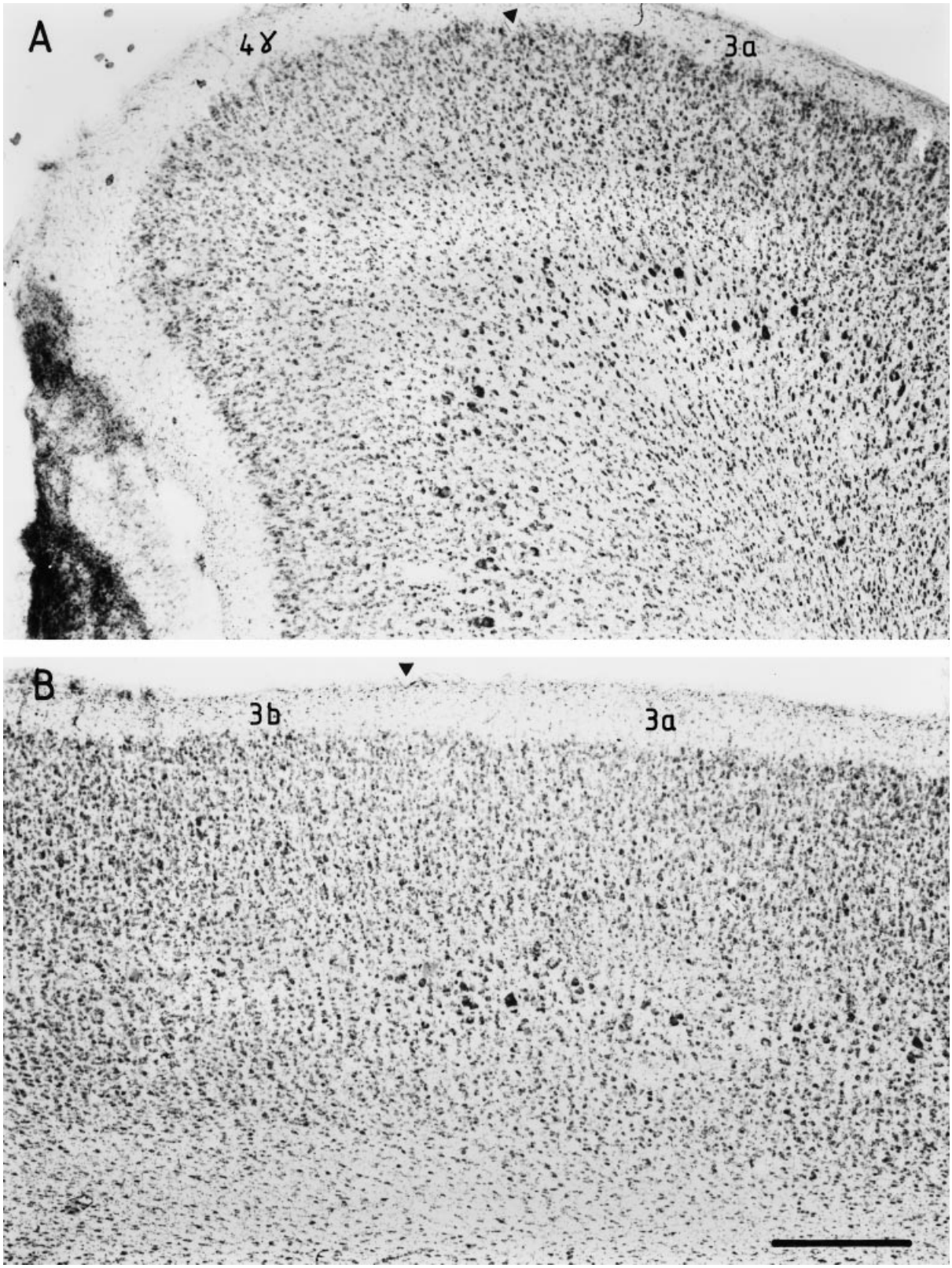


Fig. 7. Photomicrographs of coronal sections of the cat cerebral cortex, stained with cresyl violet, contrasting the cytoarchitecture of areas 4γ and 3a (**A;** the border between 4γ, left, and 3a, right, is indicated by the arrowhead above lamina II), and areas 3b and 3a (**B;** the border between 3b, left, and 3a, right, is indicated by the arrowhead above the pia), in the posterior sigmoid gyrus. Dorsal is to the left and medial is down in A, while dorsal is up and medial is to the left in B. Calibration bar = 0.5 mm.

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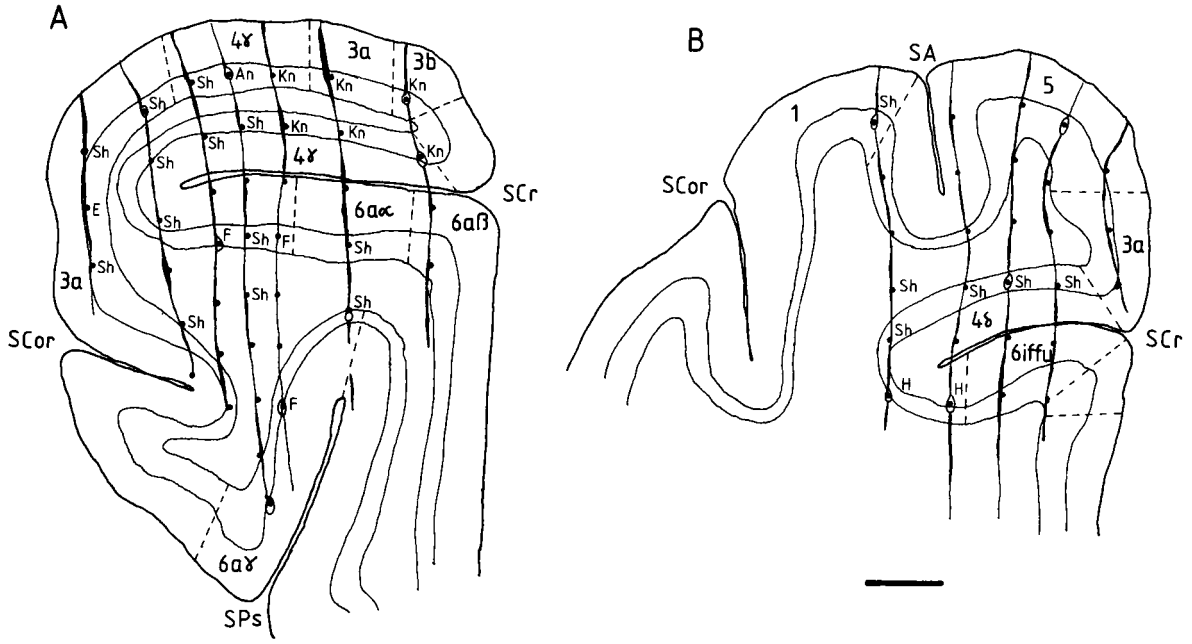


Fig. 8. **A, B:** Camera lucida drawings of two representative coronal sections of the cortex from case 2 illustrating the reconstruction of electrode tracts and stimulated points. The pial surface, the upper border of lamina V and the cortex-white matter boundary, are shown. Each row of tracts was reconstructed from three adjacent cresyl violet-stained sections. Lesions are shown as circular outlines, whereas estimated stimulus points shown as large dots. The movements evoked from these points are indicated adjacent to the point. Points without adjacent lettering were unsuccessful. In this animal, stimuli were delivered at depth intervals of 1.5 mm. Cytoarchitectonic borders are indicated by dashed lines. Calibration bar = 2 mm.

forelimb movements frequently and facial movements occasionally. Hindlimb movements were not elicited from area 6. Separate representations of forelimb and hindlimb movements in r4γ and c4γ are suggested by the data in Figures 9–11, with the forelimb representation located more laterally than the hindlimb. In one animal, facial movements were elicited from area r4γ (in the rostral bank of the cruciate sulcus adjoining area 6aα). Facial movements were not elicited from c4γ.

The caudal or dorsal bank of the cruciate sulcus includes area c4γ superficially, while area c4δ is located in its middle third. Hindlimb movements were evoked medially from area c4γ in the caudal bank of the cruciate sulcus. At the lateral edge of the cruciate sulcus, forelimb movements were evoked superficially in the sulcus and this representation was continuous with the representation of forelimb movements in the lateral and posterior sigmoid gyri. However, more deeply and medially in the cruciate sulcus, stimulation of some points in area c4γ evoked forelimb movements, intermingled with points that evoked hindlimb movements. Deeper in the cruciate sulcus, stimulation of points in area c4δ elicited forelimb and hindlimb movements. These results suggest that there is a separate representation of the forelimb and hindlimb in the deeper parts of the cruciate sulcus involving area c4δ and the adjacent part of area c4γ. However, the representation of the forelimb and hindlimb in this part of the cortex appears to be intermingled or interdigitating with no clear topographical relationship between the two. Forelimb and hindlimb movements could also be elicited from r4δ, but the points evoking these movements were closely intermingled.

Forelimb movements and more medially, facial movements were elicited from area 6aα, in the anterior sigmoid gyrus and rostral bank of the cruciate sulcus. Forelimb movements were also elicited from the dorsal lip and dorsal bank of the presylvian sulcus, in area 6aγ. Facial movements were also evoked in area 6aγ rostral to forelimb movements.

Stimulus thresholds in areas 4 and 6

The stimulus currents used for evoking movements varied from 0 to 100 μA. The threshold currents required in the different areas are compared in Table 2 and Figure 12. The mean thresholds were found to be lower in area 4γ than in areas 4δ, 6aα, 6aβ, and 3a. However, relatively low thresholds were also observed occasionally in area 4δ (case 3), 6aα (case 2) or 6aγ (case 3). Statistical analysis (ANOVA) showed that there were no significant differences ($P > 0.05$) in the mean thresholds obtained in areas 4δ, 6aα, 6aβ, and area 3a, but the mean threshold was found to be significantly lower in area 4γ ($P < 0.05$) in two animals (cases 1 and 4). Mean threshold was always lower in area r4γ than c4γ but statistically significant in only one animal (case 2, unpaired t-test, $P < 0.01$). Thresholds varied from 5 to 100 μA in most areas and a bimodal distribution of threshold currents was found in area 6aα, suggesting some very low and other very high threshold points (Fig. 12). Figure 11 suggests that the low threshold points were located medially in area 6aα close to its border with 4γ.

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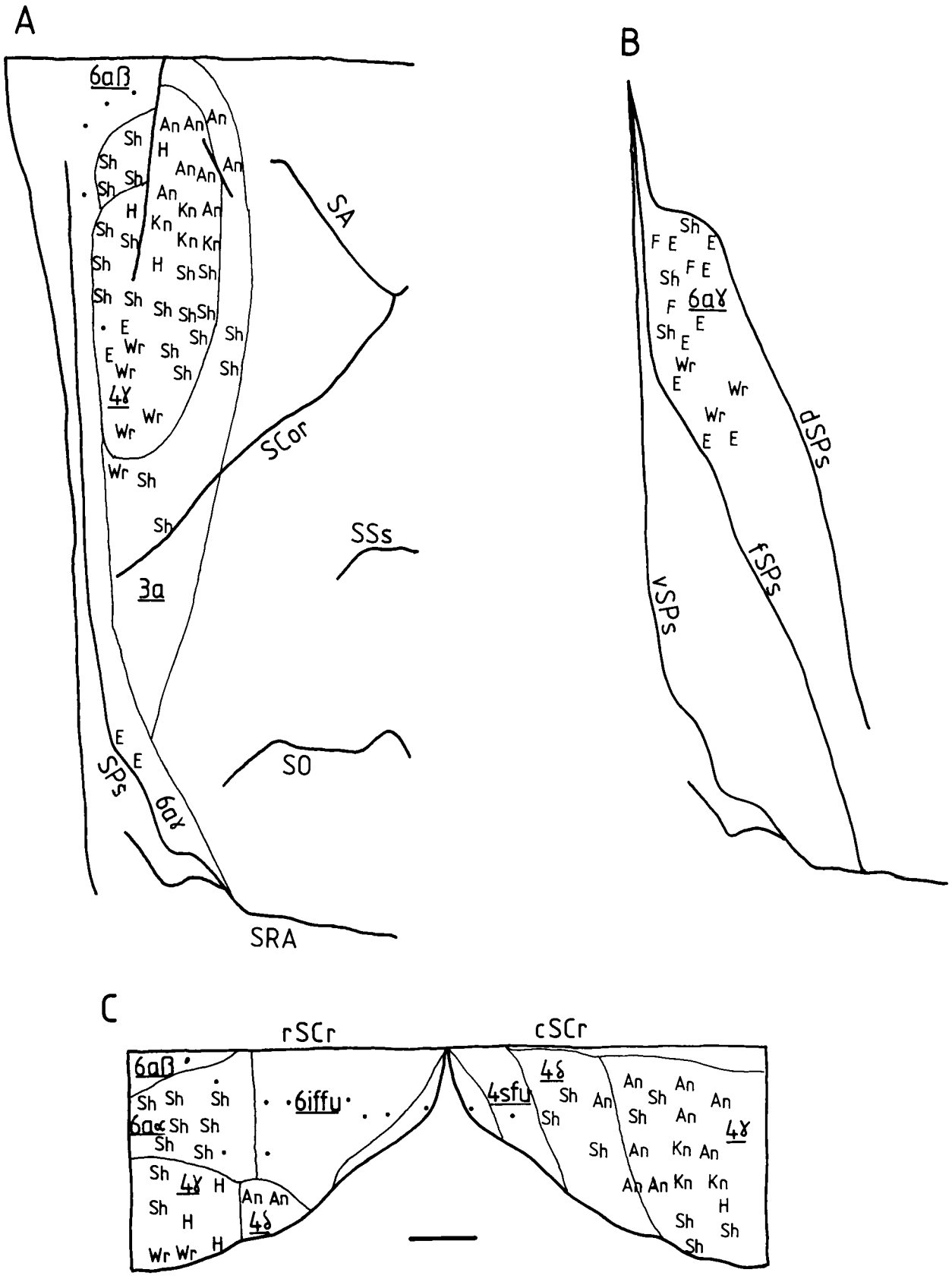


Fig. 9. Map of the dorsolateral cortical surface (**A**) and of the unfolded presylvian (**B**) and cruciate sulci (**C**), illustrating the results of intracortical microstimulation of the left hemisphere of case 1. The dorsolateral surface of the cortex has been mapped from serial coronal sections, in the plane of the upper border of lamina V, using the medial edge of the hemisphere as a reference. The presylvian sulcus was unfolded in the plane of the upper border of lamina V by reflecting the dorsal bank upwards and the ventral bank downwards. The cruciate sulcus was unfolded by reflecting the ventral (rostral) bank to the left and the dorsal (caudal) bank to the right. Contralateral joints moved by intracortical microstimulation have been depicted on the maps from reconstruction of electrode tracts and lesions in the serial sections. Calibration bar = 2 mm.

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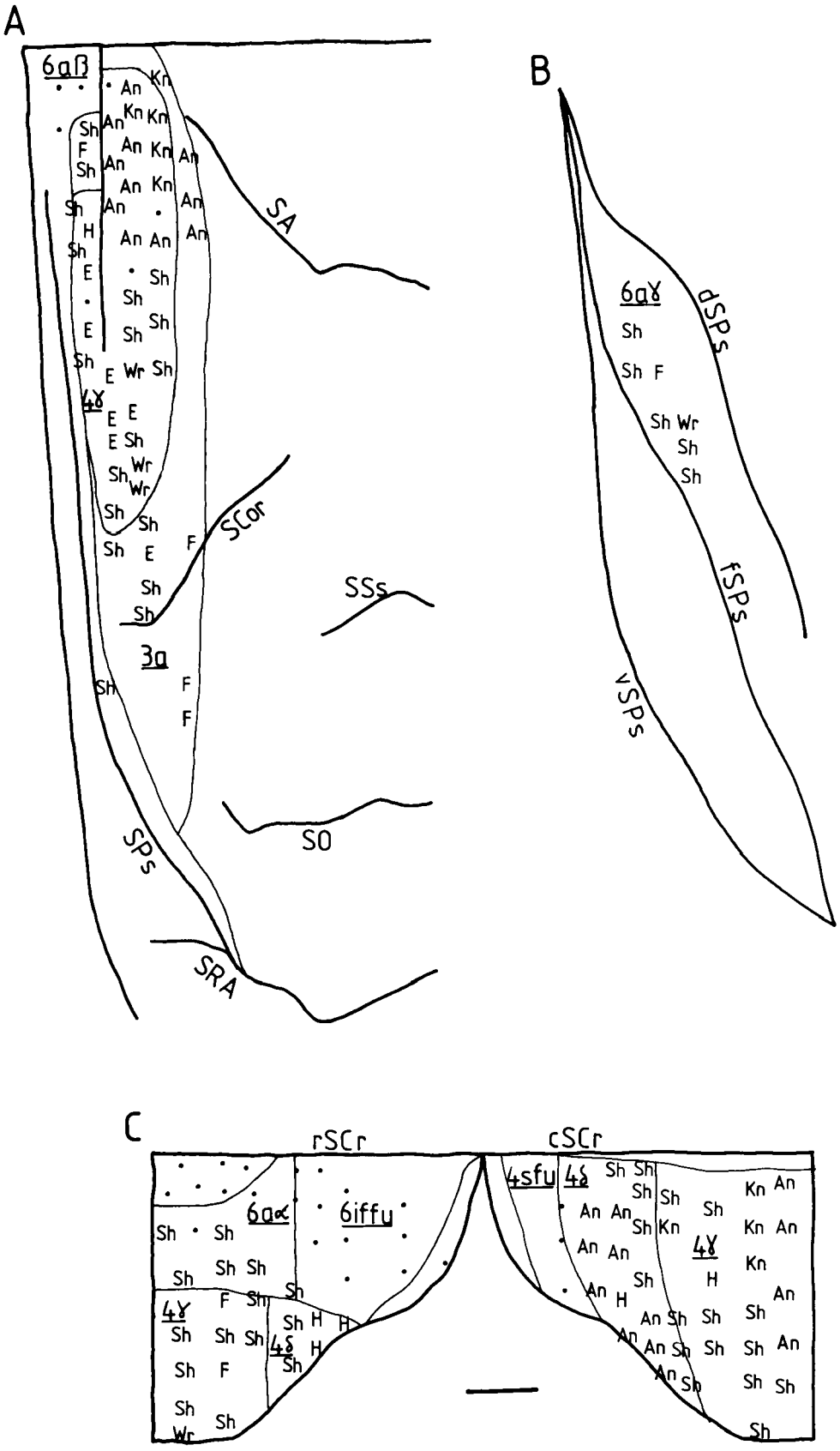


Fig. 10. Map of the dorsolateral cortical surface (**A**) and of the unfolded presylvian (**B**) and cruciate sulci (**C**), illustrating the results of intracortical microstimulation of the left hemisphere of case 2. See legend to Figure 9 for mapping techniques used. Calibration bar = 2 mm.

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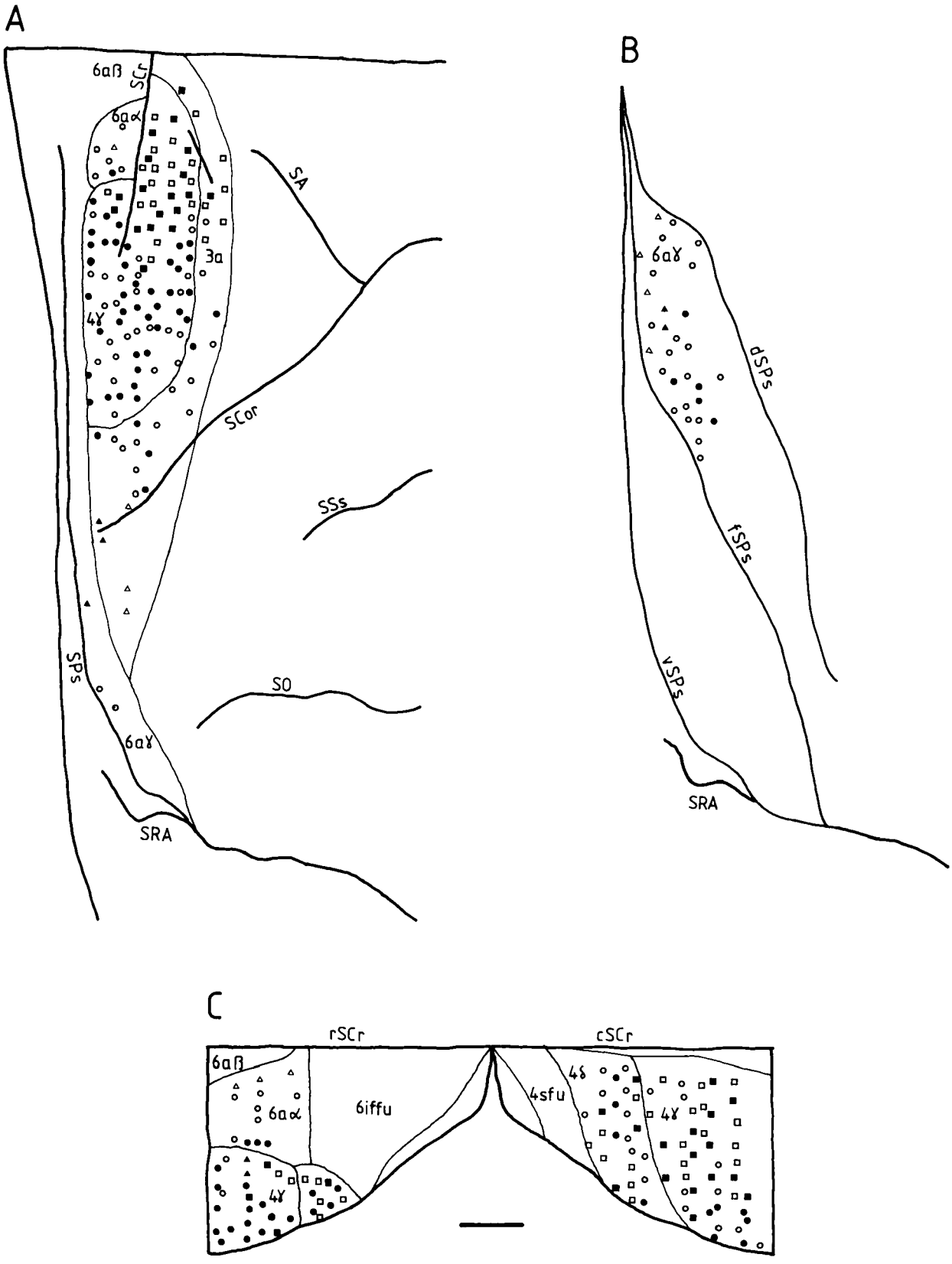


Fig. 11. Map of the dorsolateral cortical surface (**A**) and of the unfolded presylvian (**B**) and cruciate sulci (**C**), illustrating the pooled results of intracortical microstimulation of the left hemispheres of cases 1–5. See Figure 9 legend for mapping techniques used. Sites from which movements were evoked in the contralateral forelimb are represented by circles, in the hindlimb by squares, and in the face by triangles. Filled symbols indicate that currents less than 50 μA were required to evoke movements, whereas open symbols indicate the use of currents between 50 and 100 μA. Calibration bar = 2 mm.

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TABLE 2. Mean ± S.E. Intracortical Microstimulation (ICMS) Thresholds (μA) for Evoking Forelimb Movements from Different Motor Areas in Individual Animals

Animal= (case no.)	r4γ	c4γ	4γ	4δ	6aα	6aγ	3a
1	31.0 ± 7.2	50.9 ± 7.2	43.5 ± 5.5[2]	93.3 ± 6.7	59.1 ± 11.6	73.6 ± 7.5	54.0 ± 18.1
(n)	(10)	(17)	(27)	(3)	(11)	(14)	(5)
2	15.8 ± 3.9[1]	55.0 ± 7.3	41.6 ± 5.8	50.9 ± 11.0	36.9 ± 14.7	75.7 ± 15.7	76.0 ± 16.9
(n)	(13)	(25)	(38)	(11)	(8)	(7)	(5)
3	17.0 ± 6.2	25.0 ± 8.4	22.3 ± 5.9	43.9 ± 13.0	80.0 ± 20.0	43.3 ± 23.3	75.0 ± 25.0
(n)	(5)	(10)	(15)	(9)	(2)	(3)	(2)
4	17.5 ± 3.1	22.5 ± 4.2	20.8 ± 3.0[2]	66.3 ± 11.9	—	50.0 ± 0	50.0 ± 0
(n)	(6)	(12)	(18)	(8)		(2)	(2)
5	50.0 ± 14.7	55.0 ± 13.4	53.0 ± 9.4	63.3 ± 13.1	—	—	—
(n)	(4)	(6)	(10)	(6)			

[1]Mean threshold was significantly lower in r4γ than c4γ.
[2]Mean threshold was significantly lower in 4γ than other areas.

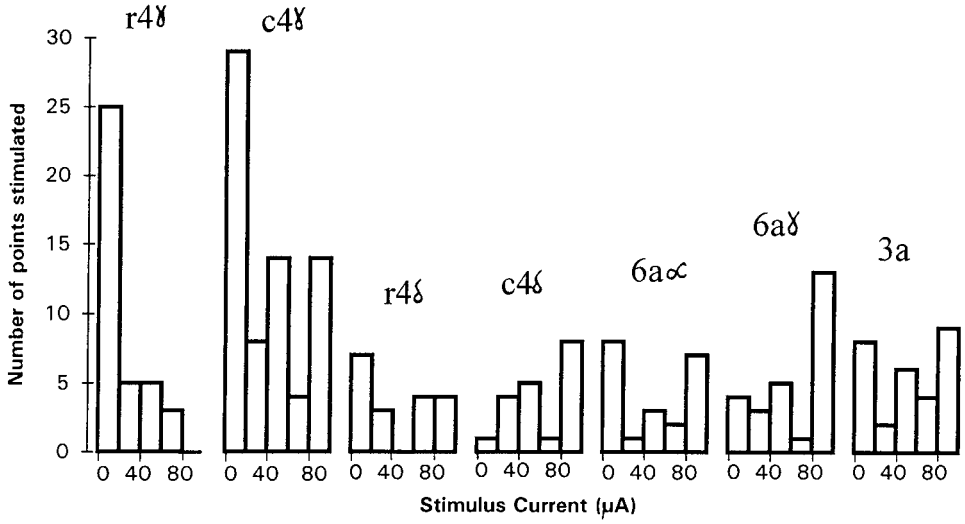


Fig. 12. Histograms comparing minimal stimulus currents required to evoke movements in the contralateral forelimb following intracortical microstimulation of cortical areas 4γ, 4δ, 6aα, 6aγ and 3a. Data from all five animals studied were pooled.

Tracer injections in the spinal cord

Fluorescent tracer injections result in three concentric zones of labeling: an inner zone with necrotic material and bright tissue fluorescence; a middle zone of bright tissue fluorescence and labeled neurons and glia; and an outer zone with no tissue fluorescence and less intensely labeled neurons and glia (Fig. 1, Keizer et al., 1983). The two inner zones were taken as constituting the zone of uptake. The single tracer injection in case 6 involved a large part of the lateral funiculus and included most if not all of the lateral corticospinal tract (Fig. 1; Nyberg-Hansen and Brodal, 1963; Armand et al., 1985). The tracer injections in cases 7–9 involved part of the dorsal columns, a large part of the intermediate zone of the gray matter, and a variable part of the dorsal and ventral horn (Figs. 1 and 13). The tracer injection never spread into the lateral funiculus but occasionally came close to the anterior funiculus (Figs. 1 and 13). However, we felt that involvement of the ventral corticospinal tract was unlikely. Each injection involved about 2–3 mm of the spinal cord rostrocaudally. However, there were segments of the cord between penetrations that were not involved by the tracer. Close to the injection site, labeled neurons were observed profusely throughout the ipsilateral gray matter and less profusely in the contralateral intermediate zone and ventral horn. Motoneurons were well labeled bilaterally.

Corticospinal neurons

Neurons labeled following tracer injections in the lateral funiculus or gray matter of the spinal cord are mapped in Figures 14–16. The mapping procedure is similar to that adopted for presentation of ICMS data. All corticospinal neurons were located in lamina V of the cortex and labeled in clusters (Groos et al., 1976; Jones and Wise, 1977; Murray and Coulter, 1981; Hayes and Rustioni, 1981; Nudo and Masterton, 1990). In all the animals studied, motor areas 4γ, 4δ, 6aα, and 6aγ, and somatosensory areas 3a, 3b, 1, 2, 2pri, and 5 were well labeled. A few cells were located in the insula. Labeled cells were rarely located in areas 6aβ, 6iffu, 4sfu, or 4fu. In the animal (case 9) with dual tracer injections in the cervical and lumbar segments of the cord, double labeled neurons were rarely seen (Fig. 15).

Dense labeling was observed in areas 4γ and in area 4δ in all animals. Labeling in area 6 was observed only following tracer injection in the cervical cord, in the gray matter or in the lateral funiculus. Labeled neurons were distributed throughout area 4γ, but parts of area 4δ, 6aα and 6aγ were unlabeled in all the four animals (Figs. 14–16). The medial part of area 6aα and r4δ, the deeper parts of c4δ, and parts of area 6aγ were consistently unlabeled.

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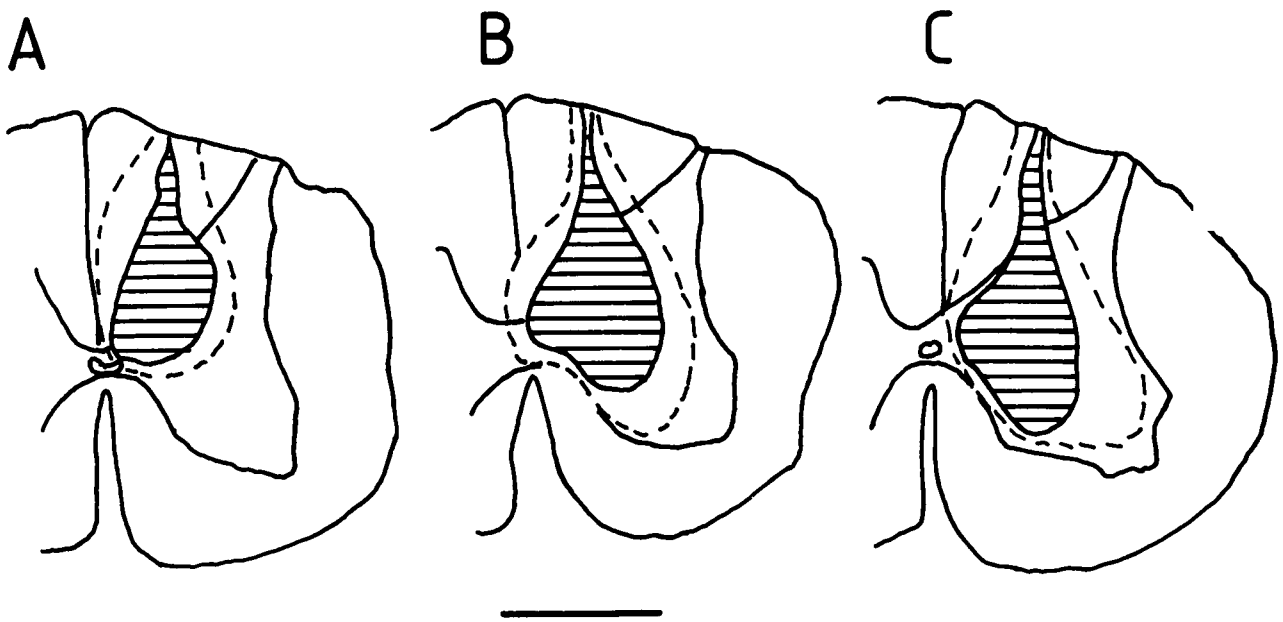


Fig. 13. Camera lucida reconstructions of one of the Diamadino Yellow injection sites in the cervical spinal cord of case 9. The sections are about 800 μm apart. Calibration bar = 2 mm.

The topographical relationship between the representation of the forelimb and hindlimb in the motor cortex can be best appreciated in the animal with dual tracer injection (case 9, Fig. 15), but additional information is also provided by cases 7 and 8 (Fig. 16). The results suggest representations of forelimb and hindlimb movements in both the rostral (r4γ) and caudal (c4γ) subdivisions of area 4γ. In c4γ, the hindlimb representation is located medial to the forelimb representation, while in r4γ, the hindlimb representation is located deeper in the cruciate sulcus and more medially in the anterior sigmoid gyrus. The forelimb representation is clearly continuous between r4γ and c4γ, and the hindlimb representation also appears to be continuous. Representations of forelimb and hindlimb movements are also seen in areas r4δ and c4δ.

Projections to the cervical spinal cord arise deep in the caudal (dorsal) bank of the cruciate sulcus involving both area c4γ and c4δ. These forelimb-related corticospinal neurons are clearly separate from those located in the posterior sigmoid gyrus (Figs. 15, 16). Thus, a separate representation of the forelimb and hindlimb in the deeper parts of the caudal bank of the cruciate sulcus (involving areas c4γ and c4δ) can easily be envisaged. It may be suggested that corticospinal neurons projecting to the cervical cord are located more superficially in the sulcus than those projecting to the lumbar spinal cord, but there is considerable interdigitation of the two populations. A few double labeled neurons located in the cortex of case 9 (Fig. 15) were found in the caudal bank of the cruciate sulcus (see Hayes and Rustioni, 1981).

Corticospinal neurons projecting to cervical segments of the spinal cord were detected in areas 6aα and 6aγ. Corticospinal neurons in area 6aγ were not spatially separate from that in area r4γ, but linked by a band of neurons located at the rostrolateral edge of r4γ and at the rostral end of area 3a (Figs. 14–16). Similarly, neuronal labeling in areas r4γ and 6aα were continuous.

Comparison of ICMS and tracer results

The data obtained from the ICMS (Figs. 9–11) and tracer (Figs. 14–16) studies were complementary. ICMS was successful only in those motor areas which subsequently were shown to possess corticospinal neurons. Similarly, representations of forelimb and hindlimb movements in r4γ, c4γ, r4δ, and c4δ, and the representation of forelimb movements in areas 6aα and 6aγ, are suggested by both sets of data.

DISCUSSION

Motor areas have been defined operationally as those regions of the cortex from which movements or brief muscle contractions are evoked with lowest threshold and/or shortest latency by applied electrical stimuli (Humphrey, 1986). It has been known for a long time that mapping of motor cortical areas using electrical stimuli, using relatively low stimulus currents, is dependant greatly upon direct corticospinal projections (reviewed by Phillips and Porter, 1977; Kuypers, 1982; Humphrey, 1986; Porter and Lemon, 1993). Therefore, it may be expected that a combination of approaches, utilizing electrical stimulation of the cortex and neuroanatomical tracing of corticospinal connections, would provide mutually complementary results and that such an approach is particularly valuable in defining motor areas in animals that lack a significant corticomotoneuronal connection (e.g., the cat, Lundberg and Voorhoeve, 1962). This study has therefore utilized the combined approach to demonstrate multiple representations of the contralateral limbs in the motor cortex of the cat. A similar approach has been used to define the borders of the primary motor area in macaques (Sessle and Wiesendanger, 1982).

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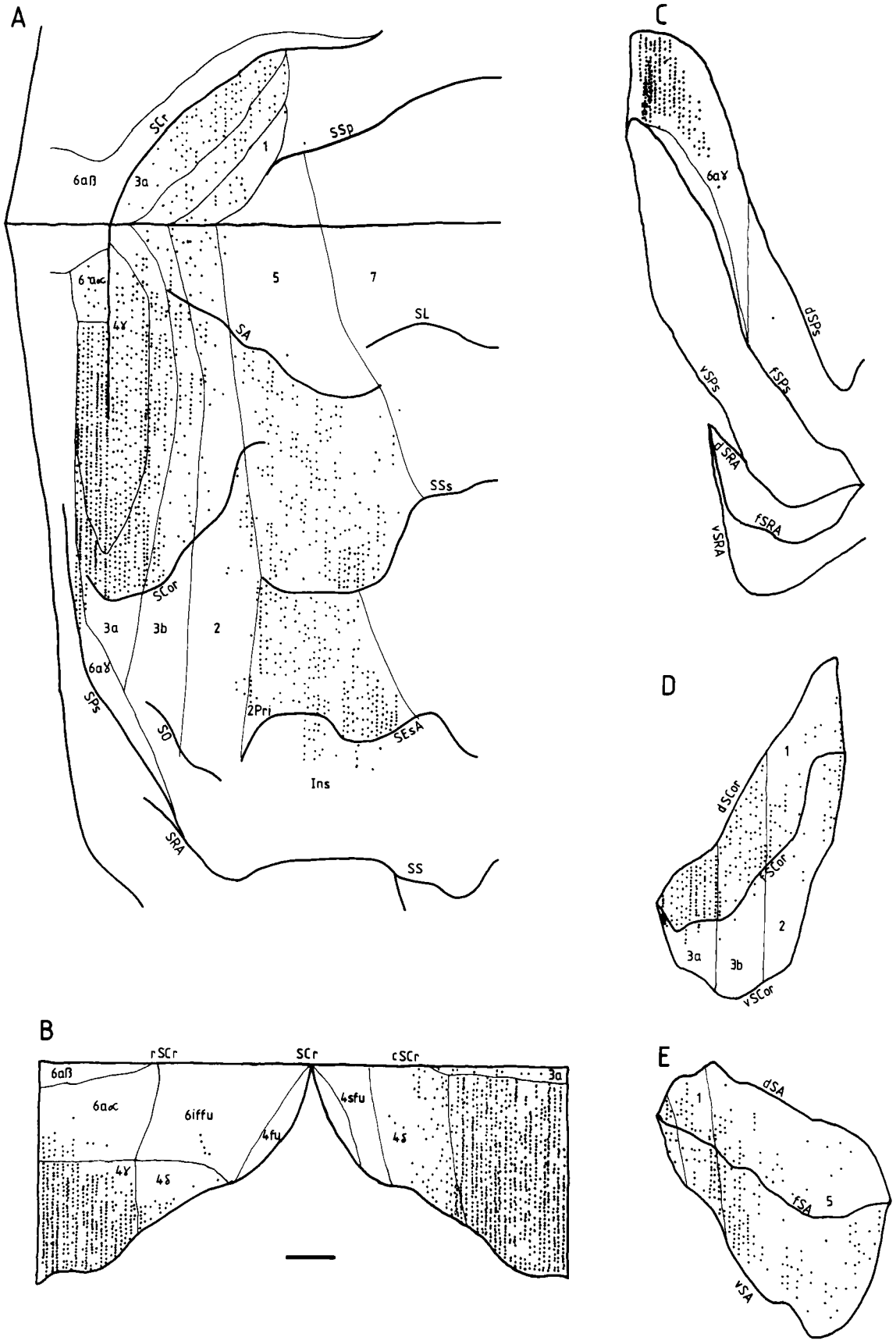


Fig. 14. Map of labeled corticospinal neurons in the medial and dorsolateral surface of the left hemisphere (**A**) and the banks of the cruciate sulcus (**B**), presylvian sulcus (**C**), coronal sulcus (**D**), and ansate sulcus (**E**) of case 6. See legend to Figure 9 for mapping techniques used. Labeled neuronal profiles were counted in every fourth section (each 50-μm-thick) of the series and plotted in 1-mm bins within the map. Each dot represents one labeled profile. Calibration bar = 2 mm.

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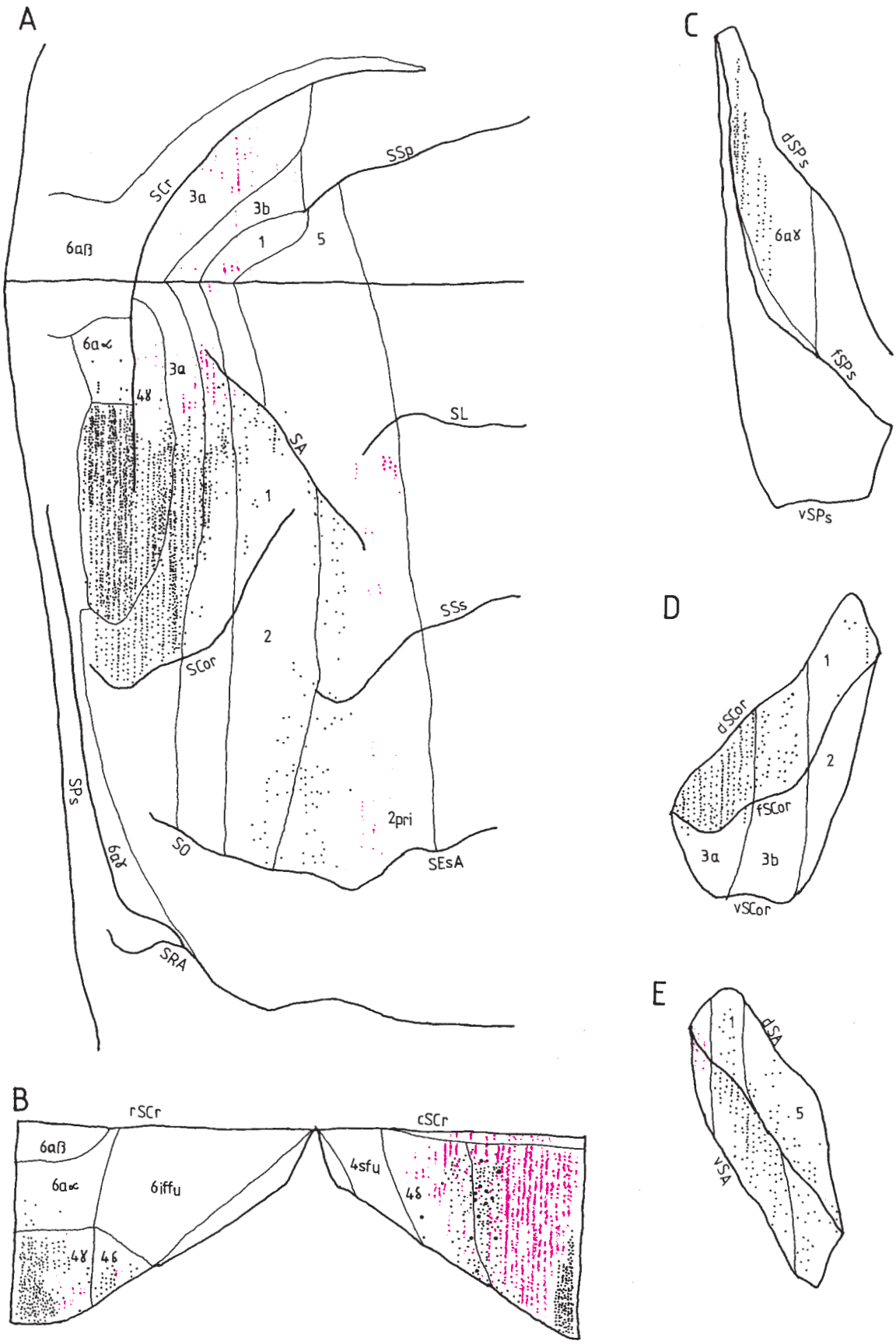


Fig. 15. Map of labeled corticospinal neurons in the medial and dorsolateral surface of the left hemisphere (**A**) and the banks of the cruciate sulcus (**B**), presylvian sulcus (**C**), coronal sulcus (**D**), and ansate sulcus (**E**) of case 9. See legend to Figure 9 for mapping techniques used. Labeled neuronal profiles were counted in every fourth section (each 50-μm-thick) of the series and plotted in 1-mm bins within the map. Each dot represents one labeled profile; red dots represent neurons labeled by Fast Blue (injected in the lumbar segments of the spinal cord), small black dots represent neurons labeled by Diamidino Yellow (injected in the cervical spinal cord), and large black dots represent double labeled neurons. Calibration bar = 2 mm.

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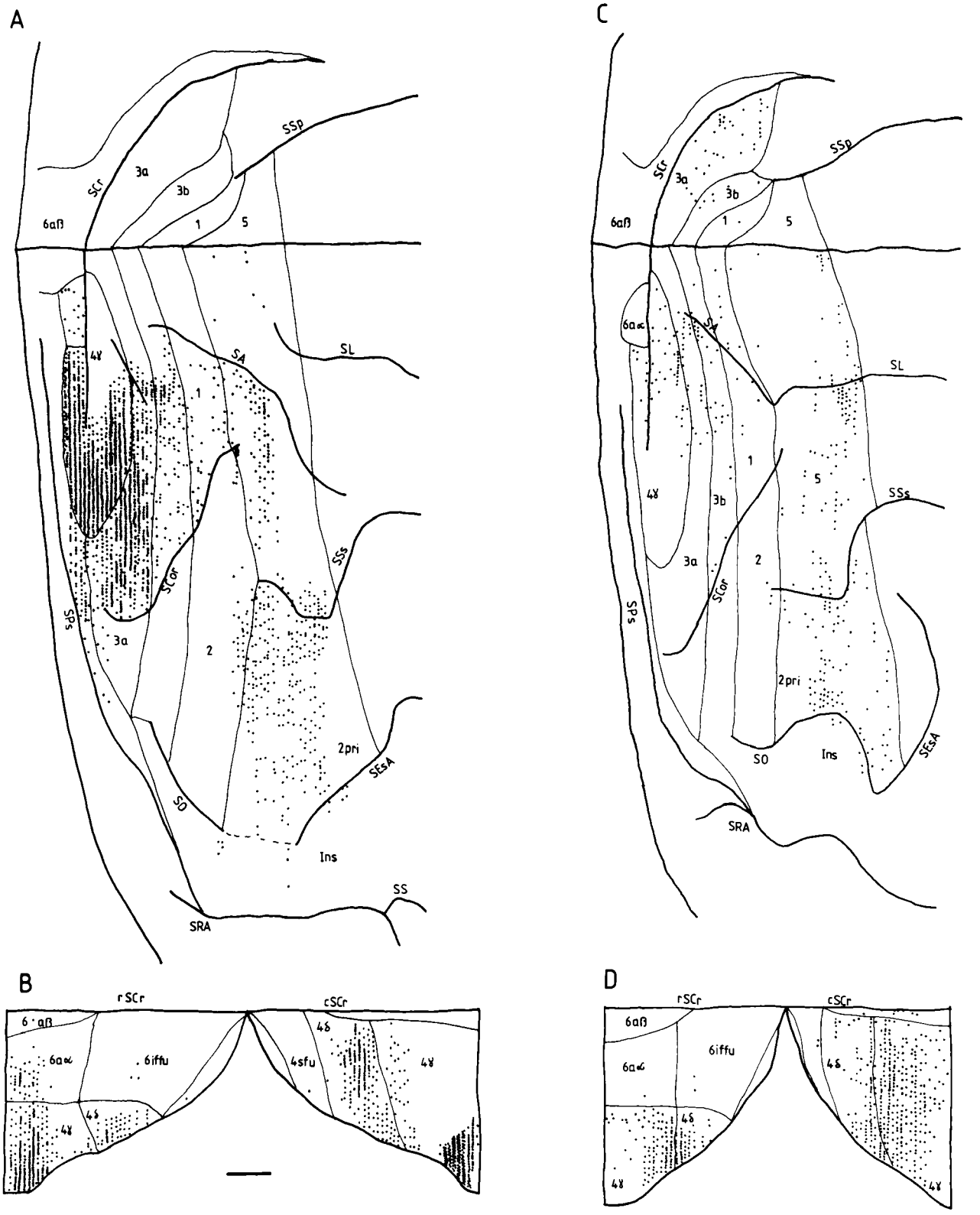


Fig. 16. Map of labeled corticospinal neurons in the medial and dorsolateral surface of the left hemisphere (**A, C**) and the banks of the cruciate sulcus (**B, D**) of cases 7 (A, B, tracer injected in cervical segments of the spinal cord) and 8 (C, D, tracer injected in lumbar segments of the spinal cord). See legend to Figure 9 for mapping techniques used. Labeled neuronal profiles were counted in every fourth section (each 50-μm-thick) of the series and plotted in 1-mm bins within the map. Each dot represents one labeled profile. Calibration bar = 2 mm.

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Cortical mapping by electrical stimulation

Electrical stimulation spreads physically and physiologically (Asanuma and Rosen, 1973; Jankowska et al., 1975; Asanuma et al., 1976; Phillips and Porter, 1977) and therefore cannot always resolve the exact limits of the "excitable" cortex, nor clearly define topographical representation of movement where there is considerable overlap or interdigitation of the representation of different body parts. The use of large currents for ICMS may provide misleading results. Nevertheless, electrical stimulation studies, particularly in the macaque monkey (using a similar range of currents), have provided information on cortical topography not only in the MI, but also in the PM, SMA and the CMA (Asanuma and Rosen, 1972; Anderson et al., 1975; Kwan et al., 1978; MacPherson et al., 1982; Mitz and Wise, 1987; Rizzolatti et al., 1988; Luppino et al., 1991). The presence of corticomotoneuronal connections in primates, particularly involving distal limb muscles (reviewed by Porter and Lemon, 1993), allows the use of smaller currents for evoking movements and a greater resolution in cortical cartography. However, even in the cat, relatively small currents (<20 μA) were effective in a proportion of the points stimulated in each of the "excitable" areas.

The results of our microstimulation study agree with the previous findings of Keller (1993), Nieoullon and Rispal-Padel (1976), Delgado (1952) and Delgado and Livimgston (1955). However, in our study, movements of facial muscles were evoked infrequently and jaw and back muscles could never be activated. In addition, retraction of claws was also not evoked. These differences may in part be due to the differences in the preparation, including the state of consciousness and depth of anaesthesia, type of anaesthetic used, the techniques utilized to monitor muscle contraction (e.g., visual observation, e.m.g. recording or transducer techniques), electrode resistance and stimulus currents (Delgado, 1952; Schlag and Schlag-Rey, 1970; Asanuma and Ward, 1971; Nieoullon and Rispal-Padel, 1976; Pappas and Strick, 1981; Iwata et al., 1990; Keller, 1993). Some studies (e.g., Pappas and Strick, 1981; Keller, 1993) that have made a more detailed study of a part of area 4γ (involving mainly or exclusively the forelimb representation), have also been able to detect many more low threshold points for evoking movements of forelimb and facial muscles.

Our study found representation of limb movements in several cytoarchitectonic subdivisions of the motor cortex: areas 4γ, 4δ, 6aα, and 6aγ. While there was considerable overlap of stimulus currents required to elicit movements from these areas, the mean thresholds were lowest in area 4γ and higher in areas 4δ, 6aα and 6aγ.

Corticospinal projections in the cat

Corticospinal projections were investigated using tracer injections in the lateral funiculus or intermediate gray matter of the spinal cord. In both instances, it is unlikely that all corticospinal axons or their terminals were accessed. The tracer injection in the lateral funciculus was located in the region of the lateral corticospinal tract (Armand et al., 1985). Damage to the lateral corticospinal tract was evident, so that the tracer (Fast Blue) was taken up by both damaged and undamaged axons (Payne, 1987). Fibers descending in the ventral corticospinal tract, however, were not sampled (Armand and Kuypers, 1980). It has been estimated that the pyramidal tract of the cat contains about 400,000 fibers (both myelinated and unmyelinated, Thomas et al., 1984) and the total (estimated) number of neuronal profiles labeled (in the contralateral hemisphere) in our experiments varied from 14,000 to 16,000. Whereas not all fibers of the pyramidal tract neurons descend into the spinal cord, it would still appear that less than 10% of the corticospinal neurons were labeled in each experiment.

Tracer injections were directed to the intermediate zone of the gray matter in order to involve corticospinal terminations, particularly from the motor areas (Nyberg-Hansen and Brodal, 1963; Armand et al., 1985). Other studies have utilized this technique to investigate corticospinal projections (Groos et al., 1978; Hayes and Rustioni, 1981; He et al., 1993, 1995). Whereas a large part of the intermediate zone was involved in the region of tracer injections, the involvement of the dorsal and ventral horn was usually small and variable. Therefore, the proportion of corticospinal neurons labeled in the somatosensory areas may be relatively smaller than in the motor area because the projections from the former area terminate mainly in the dorsal horn (Nyberg-Hansen and Brodal, 1963; Armand et al., 1985).

The use of multiple retrograde tracers to investigate the topography of descending projections provides many advantages (see Catsman-Berrevoets and Kuypers, 1981; Hayes and Rustioni, 1981; He et al., 1993, 1995). In this study, the descending projections to cervical and lumbar segments of the spinal cord were compared in one animal using two tracers, and in two other animals, tracers were injected either in the cervical or lumbar cord. It was important to ensure that decending tracts were not involved in the tracer injections at the cervical levels; hence, penetrations were made through the dorsal columns (Groos et al., 1978; Hayes and Rustioni, 1981; He et al., 1993, 1995) and the injections made at depths that precluded involvement of the ventral funiculus.

Corticospinal projections were found to arise from a number of agranular regions. Although the bulk of the cortical labeling was located in area 4γ, a number of neurons were also labeled in areas 4δ, 6aα and 6aγ. These findings agree with other reports of the location of corticospinal neurons in area 6 (Catsman-Berrevoets and Kuypers, 1981; Hayes and Rustioni, 1981; Nudo and Masterton, 1990). In our study, labeling in area 6aα was sparse and that in area 6aγ was moderate, suggesting that contribution to the corticospinal tract from area 6 is much smaller than from area 4.

Corticospinal neurons projecting to both cervical and lumbar segments were rare. Whereas individual corticospinal axons arborize extensively in the rostrocaudal dimensions of the spinal cord (Shinoda et al., 1976, 1986), few appear to project to both cervical and lumbar segments (Shinoda et al., 1976, 1986; Hayes and Rustioni, 1981).

This study provides evidence for at least three representations of the contralateral limb(s) outside area 4γ. One of these is located in the depth of the caudal bank of the cruciate sulcus, involving area c4δ and the adjacent part of c4γ. The representation of forelimb and hindlimb movements was also detected in r4δ. In these regions, there was considerable interdigitation of forelimb and hindlimb related areas (see discussion in Groos et al., 1978). A second representation of the contralateral forelimb was detected in area 6aγ, in the dorsal bank of the presylvian sulcus. A

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smaller region of cortex in area 6aα was also found to be involved in the control of forelimb movements. The representation of limb movements in area 4δ appeared to be spatially separate from that in area 4γ, but the representations of forelimb movements in areas 6aα and 6aγ were in continuity with that observed in r4γ.

Motor areas in the cat

Many studies have examined the representation of movement in the motor cortex of the cat (MacKibben and Wheelis, 1932; Ward and Clark, 1935; Delgado, 1952; Delgado and Livingston, 1955; Woolsey, 1964; Asanuma and Sakata, 1967; Asanuma et al., 1968; Asanuma and Ward, 1971; Nieoullon and Rispal-Padel, 1976; Pappas and Strick, 1981; Keller, 1993) and provided a detailed account of the topography of movement representation in the primary motor area (Delgado, 1952; Delgado and Livingston, 1955; Nieoullon and Rispal-Padel, 1976; Keller, 1993). Nieoullon and Rispal-Padel (1976) used ICMS in cats chronically implanted with electrodes to reveal a detailed representation of the contralateral head, neck, face, back, forelimb, and hindlimb muscles. Their studies suggested a single representation of these structures with considerable overlap or interdigitation, and involving area 4γ, area 6aα, and possibly area 6aγ (see Fig. 5 of their study). In their scheme, hindlimb movements were represented mainly in the posterior sigmoid gyrus and in the caudal bank of the cruciate sulcus, forelimb movements in the lateral sigmoid gyrus and adjoining parts of the anterior and posterior sigmoid gyri, and face and head movements in areas 6aα and 6aγ. Although a part of the caudal (or dorsal) bank of the cruciate sulcus was explored, many other regions in the banks of the cruciate sulcus or the presylvian sulcus were not examined. Keller's (1993) study involved area 4γ, and possibly area 6aγ, and suggested that the representation of facial muscles is located lateral to that of forelimb muscles. Delgado (1952) and Delgado and Livingston's (1955) studies found a representation of cranial and forelimb muscles in the dorsal bank of the presylvian sulcus. Whereas there are suggestions that certain movements or muscles may be represented more than once in the motor cortex (e.g., Pappas and Strick, 1981; Keller, 1993), all previous studies have implied an overall single representation of contralateral body parts in the motor cortex of the cat.

Studies of the representation of movement in other carnivores (racoons and dogs), on the other hand, have suggested a dual representation of the body in the motor cortex, separated by the cruciate sulcus: MI being located caudal to the SMA (Hardin et al., 1968; Jameson et al., 1968; Gorska, 1974). In these studies, current thresholds were found to be higher in the SMA, and more complex and bilateral movements were evoked from this area.

This study found representations of both contralateral limbs, using both ICMS and tracer studies, in two cytoarchitectonic areas of the agranular cortex: area 4γ and area 4δ. In addition, the forelimb was found to be represented in areas 6aα and 6aγ. Further, representation of forelimb and hindlimb muscles was detected in both rostral and caudal subdivisions of 4γ and 4δ. Facial muscles were found to be represented in areas 6aα, 6aγ and in a small part of area 4γ. Whereas these findings are in broad agreement with previous results, they also suggest multiple representations of contralateral body parts in the motor cortex of the cat.

For a number of reasons, area 4γ has been considered to be the primary motor area in the cat. These include cytoarchitectonic criteria, low current requirements for evoking movements by cortical stimulation, and the presence of large numbers of corticospinal neurons. Nevertheless, while most studies have found large regions of 4γ involved in controlling limb movements, many studies (including ours) found it difficult to localize a face area in 4γ. Labeled corticospinal neurons were found to involve practically the whole extent of area 4γ, leaving free only a small part adjacent to area 6aα, from which facial movements were occasionally evoked. It appears that a substantial part of the representation of facial movements in cat motor cortex lies outside area 4γ. Previous studies have included the representation of facial muscles in either area 6aα, 6aγ, or both, within MI (Nieoullon and Rispal-Padel, 1976; Keller, 1993). The present study found that facial movements could be evoked with relatively small currents ($<$50 μA) from a few points within area 6aγ, whereas current requirements appeared to be higher in area 6aα (always $>$ 50 μA, Fig. 11). In addition, Keller (1993), Iwata et al. (1990) and Nieoullon and Rispal-Padel's (1976) studies also appear to identify area 6aγ as the low threshold site for evoking facial movements. Thus, one scheme of the topographic organization in cat MI could envisage a mediolaterally oriented representation of hindlimb, forelimb and facial movements, respectively, and including areas 4γ and 6aγ.

A representation of forelimb muscles was also detected in the dorsal (or lateral) lip and bank of the presylvian sulcus. This representation of forelimb muscles in area 6aγ appears to involve both proximal and distal musculature and is the site of origin of large numbers of corticospinal neurons. Area 6aγ has been considered a part of the primary MI by previous studies, because a representation of facial muscles has been difficult to detect in the lateral parts of area 4γ. Nevertheless, cytoarchitectural and connectional differences (see below), and differences in currents required for evoking movements suggest an alternative possibility: that the representation of forelimb movements in area 6aγ may be separate from that in area 4γ. The representation of facial muscles in area 6aγ is also not homogeneous. Iwata et al. (1990) distinguish two separate representations: one in anterior part of the coronal and lateral sigmoid gyri (C-S motor zone) and another in the dorsal (lateral) bank of the presylvian sulcus (P motor zone) on the basis of ICMS effects and responses to peripheral stimulation. Thus, if the C-S zone is considered to be part of MI, then a second representation of forelimb and facial muscles (P-zone) can be envisaged in the dorsal (lateral) bank of the presylvian sulcus. This region could possibly be a homologue of the primate PMv (see below).

A second interpretation of results could visualize the representation of facial muscles in the medial edge of r4γ (and possibly including that in area 6aα) as part of MI, with a complete body map folded across the cruciate sulcus (Nieoullon and Rispal-Padel, 1976). Further studies may be required to indicate whether representations of forelimb and facial movements in area 6aα and 6aγ are part of or separate from the primary motor cortical representation of these movements.

The supplementary motor area has never been clearly defined in the cat. However, a number of studies have pointed out differences in the connections and functions of

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rostral and caudal subdivisions of area 4γ, separated by the cruciate sulcus (Adkins et al., 1971; Vicario et al., 1983; Yumiya and Ghez, 1984). Because, in other carnivores, the SMA has been tentatively identified in the anterior sigmoid gyrus (Hardin et al., 1968; Jameson et al., 1968; Gorska, 1974), SMA may also be located in the anterior sigmoid gyrus of the cat. The present study found a representation of forelimb, hindlimb and facial muscles in r4γ, and the representation of forelimb and facial muscles in 6aα. However, the somatotopic arrangement of these structures is less clear than in the posterior sigmoid gyrus. A large area of cortex, representing forelimb muscles, occupies most of r4γ; very small regions representing hindlimb or facial muscles were found in the medial edge of r4γ and deep in the rostral (ventral) bank of the cruciate sulcus. The few points in r4γ, from which facial or hindlimb muscles were activated, also precludes a clear indication of the somatotopic arrangement in this area. Nevertheless, a case may be made for a representation of contralateral body parts in the anterior sigmoid gyrus in the cat, which may be a homologue of the primate SMA (see below). However, the forelimb representation in r4γ appeared to require lower currents for ICMS than c4γ (although statistically significant observations were made in only 1 of 5 animals). This finding is different from that observed in dogs and racoons (Hardin et al., 1968; Jameson et al., 1968; Gorska, 1974), although in these animals MI was found to encroach upon the anterior sigmoid gyrus, and therefore, those studies may not be strictly comparable with ours. Nevertheless, we or others have not found a clearly separate area in the cat r4γ with *higher* thresholds. Another important finding which argues against the location of an SMA in the anterior sigmoid gyrus in the cat is the scarcity of cortical connections between the anterior and posterior sigmoid gyri (Yumiya and Ghez, 1984; unpublished observations in our laboratory).

In this study, a representation of the contralateral limbs was detected in area 4δ, and may be an alternative candidate for the SMA in the cat. The current thresholds for evoking movements were higher than 4γ and large numbers of corticospinal neurons were detected in this region. The representation of the forelimb and hindlimb appeared to be closely interdigitated in this area. This interdigitation of forelimb and hindlimb areas in the caudal bank of the cruciate sulcus was also reported by Hayes and Rustioni (1981). A representation of cranial muscles was not detected in area 4δ in this study. The medial portion of r4δ, found devoid of corticospinal neurons, may contain such a representation, which was missed in this study.

Comparison with primates

The structure, function and connections of motor areas have been studied most rigorously in macaques. Based on studies of cortical architecture, stimulation, connections, lesions, and association of neural activity with motor performance, at least four major motor areas controlling limb movements have been identified in this species: MI, SMA, PM, and CMA (reviewed by Wise et al., 1991; Dum and Strick, 1991; Kurata, 1994a). These areas can be further subdivided on the basis of structural and functional differences (e.g., pre-SMA and SMA, PMv and PMd, CMAr, CMAv and CMAd; Rizzolatti et al., 1988; Luppino et al., 1991; Kurata, 1991, 1994a; He et al., 1993, 1995; Ghosh and Gattera, 1995). Many of these areas have also been identified in other primates (Gould et al., 1986; Stepniewska et al., 1993; Wise, 1996). However, inspite of the availability of detailed information in primates, there are few properties, if any, that can be used to unequivocally identify a particular motor area (see Stepniewska et al., 1993; Wise, 1996). It is also clear that boundaries between primary and nonprimary motor areas can rarely be defined by ICMS or by the mapping of corticospinal neurons alone (Sessle and Wiesendanger, 1982; Mitz and Wise, 1987; Rizzolatti et al., 1988; Nudo and Masterton, 1990; He et al., 1993, 1995).

Whereas MI is considered to be located in area 4γ of the cat, there are both similarities and differences between cat and primate MI. Similarities include cytoarchitecture, the presence of large numbers of corticospinal neurons and the requirement of relatively low currents to evoke movements by stimulation. MI of both cats and monkeys receive input predominantly from "cerebellar" thalamic territories (Jones et al., 1979; Schell and Strick, 1984; Yumiya and Ghez, 1984; Ghosh et al., 1987; Moran and Reinoso-Suarez, 1988; Inase and Tanji, 1995). There are major differences, between the two orders, in the projection and termination of corticospinal axons in the spinal cord (reviewed by Phillips and Porter, 1977). In this and previous studies, it was noted that the face representation in area 4γ is much smaller than that in primate MI (see above). In addition, separate rostral and caudal subdivisions can be identified in cat MI which do not appear to correspond to the rostral and caudal subdivisions identified in primate MI (Stepniewska et al., 1993). Firstly, ICMS thresholds appear to be lower in r4γ than in c4γ, and secondly, there are few connections, if any, between r4γ and c4γ in the cat (Yumiya and Ghez, 1984) unlike that seen in the primate (Stepniewska et al., 1993).

The SMA is located rostromedial to MI in primates, is somatotopically organized, and projects to the spinal cord and MI (reviewed by Wise, 1996). Whereas current thresholds for evoking movements are generally held to be higher in SMA, this has not always been found to be so (e.g., Mitz and Wise, 1987; see Wise, 1996). In addition, there are suggestions that thalamic territories receiving inputs from the basal ganglia project preferentially to SMA, although "cerebellar" thalamic projections to SMA have also been described (Schell and Strick, 1984; Weisendanger and Wiesendanger, 1985; Darian-Smith et al., 1990; Nakano et al., 1993). From the above descriptions, it would be difficult to consider a homologue of primate SMA in r4γ of the cat, as has been suggested in other carnivores (see above). Thresholds in r4γ appear to be lower than c4γ, and projections from r4γ to c4γ are sparse or absent in the cat. Further, although thalamic projections to r4γ and c4γ are clearly different, predominantly "cerebellar" thalamic territories project to both areas (Yumiya and Ghez, 1984; Moran and Reinoso-Suarez, 1988). It would be equally difficult to locate SMA in area 6aα, because corticospinal connections from this area are sparse, and projections from 6aα to area 4γ are few and restricted to r4γ (Yumiya and Ghez, 1984). An alternative candidate for SMA in the cat is area 4δ, which has large numbers of corticospinal neurons and a large representation of the contralateral limbs. In addition, studies in our laboratory have revealed somatotopically organized reciprocal connections between areas 4γ and 4δ (unpublished observations). However, whereas the origin of thalamic projections to area 4δ is different from that to area 4γ, predominantly "cerebellar" thalamic territories project to area 4δ (Moran and Reinoso-Suarez, 1988).

The lateral PM of primates has been located in lateral area 6, is somatotopically organized, and projects to the

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spinal cord and MI (reviewed by Wise et al., 1991; Dum and Strick, 1991; Stepniewska et al., 1993; Kurata, 1994a). PM has been subdivided into dorsal and ventral subdivisions (PMd and PMv) on the basis of anatomical and functional differences (e.g., Rizzolatti et al., 1988; Kurata, 1991, 1994b; Boussaoud and Wise, 1993; He et al., 1993; Ghosh and Gattera, 1995). PMv in macaques has a representation of forelimb and facial muscles, whereas forelimb and hindlimb representations have been identified in PMd (Kurata et al., 1985; Rizzolatti et al., 1988; He et al., 1993; Ghosh and Gattera, 1995). A face representation in the PMd of new world monkeys has also been identified (Stepniewska et al., 1993). Thalamic territories related to the cerebellum and basal ganglia have been shown to project to PM, with former regions projecting preferentially to PMv and the latter to PMd (Schell and Strick, 1984; Matelli et al., 1989; Kurata, 1994b; Inase and Tanji, 1994). In this study, a representation of forelimb muscles was detected in area 6aγ. According to the study of Moran and Reinoso-Suarez (1988), thalamic projections to area 6aγ (cases GA-216 and GA-195 of their study; these authors have not distinguished between area 6aγ and 6aβ in their study) are different from those to area 4γ. Area 6aγ receives inputs mainly from "cerebellar" thalamic territories. Area 6aγ projects to area 4γ, but this projection may be restricted to r4γ (Yumiya and Ghez, 1984). Therefore, it may be suggested that area 6aγ is homologous to the primate PMv. It is difficult to identify a homologue of PMd in the cat, although area 6aα needs to be considered as a possible candidate, because it receives a large input from the "pallido-nigral" thalamic territory (Moran and Reinoso-Suarez, 1988), but it has few corticospinal neurons and projects sparsely to area 4γ.

A CMA, comparable to that in the primate, has not been identified in the cat by this or other studies. Cingulate areas in the cat (Reinoso-Suarez, 1984; Musil and Olson, 1988; Olson and Musil, 1992) were not found to be "excitable" or to project to the spinal cord, and they do not project to the primary motor area of the cat (Musil and Olson, 1988; Olson and Musil, 1992).

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